Exercise 2: Autoencoders on Fashion-MNIST

This notebook is the submission template for Exercise 2. Fill in the cells marked `TODO` / `YOUR CODE HERE` and keep the provided function names, class names, and cell order unchanged so the notebook can be graded automatically.

You will implement:

1. a fully connected autoencoder,
2. a denoising autoencoder using the same fully connected model,
3. a convolutional autoencoder,
4. short written comparisons of the reconstruction quality.

Submission Notes

Run the notebook from top to bottom before submitting. Export a PDF version as requested in the assignment, and submit both the `.ipynb` and PDF in your ZIP file.

The training cells save the best model weights as `.pth` files. You may lower `args.epochs` while debugging, but for the final run it should be at least `50` epochs as specified in the exercise sheet.

In [1]:
```
import os
from types import SimpleNamespace

import numpy as np
import torch
import torch.nn as nn
import torch.optim as optim
from torch.utils.data import DataLoader

from torchvision import datasets, transforms
from torchvision.utils import make_grid

import matplotlib.pyplot as plt
```

In [2]:
```
%matplotlib inline
plt.rcParams['figure.figsize'] = (12.0, 6.0)
plt.rcParams['image.cmap'] = 'gray'

torch.manual_seed(42)
if torch.cuda.is_available():
    torch.cuda.manual_seed_all(42)
```

Config

These hyperparameters follow the assignment specification. Keep them unchanged for the final submission unless your tutor explicitly says otherwise.

In [3]:
```
args = SimpleNamespace()
args.device = torch.device('cuda' if torch.cuda.is_available() else 'cpu')
args.cuda = args.device.type == 'cuda'
args.test_batch_size = 10
args.batch_size = 128
args.lr = 0.001
args.weight_decay = 1e-5
args.log_interval = 100
args.epochs = 50
args.noise_factor = 0.5
args.is_noisy = False

print(f'Using device: {args.device}')
```

```
Using device: cuda
```

Helper Functions

These cells are provided. You do not need to edit them.

In [4]:
```
class UnNormalize(object):
    def __init__(self, mean, std):
        self.mean = mean
        self.std = std

    def __call__(self, tensor):
        for t, m, s in zip(tensor, self.mean, self.std):
            t.mul_(s).add_(m)
        return tensor


def to_img(x, shape=(28, 28)):
    """Map a flattened tensor from [-1, 1] to image tensors in [0, 1]."""
    x = 0.5 * (x + 1)
    x = x.clamp(0, 1)
    x = x.view(x.size(0), 1, shape[0], shape[1])
    return x


def add_white_noise(x, factor=0.5, stddev=1):
    noise = x.clone().normal_(0, stddev)
    return x + noise * factor


class AddWhiteNoise(object):
    def __init__(self, stddev=1, noise_factor=0.5):
        self.stddev = stddev
```

```
        self.factor = noise_factor

    def __call__(self, tensor):
        tensor = add_white_noise(tensor, factor=self.factor, stddev=self.std
        return tensor.clamp_(0, 1)
```

Data Preparation

Fashion-MNIST images are normalized to the range `[-1, 1]`, which is why the models use `Tanh` in their final decoder layer.

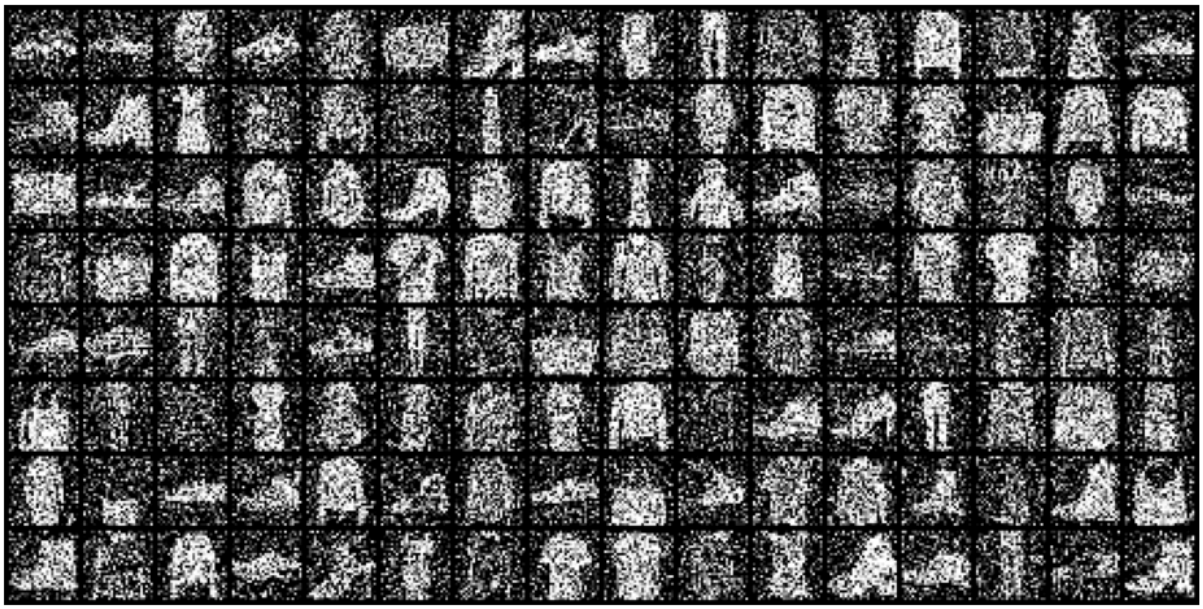
In [5]:
```
train_mean, train_std = (0.5,), (0.5,)

train_loader = DataLoader(
    datasets.FashionMNIST(
        './data_fmnist',
        train=True,
        download=True,
        transform=transforms.Compose([
            transforms.ToTensor(),
            transforms.Normalize(train_mean, train_std),
        ]),
    ),
    batch_size=args.batch_size,
    shuffle=True,
)

test_loader = DataLoader(
    datasets.FashionMNIST(
        './data_fmnist',
        train=False,
        download=True,
        transform=transforms.Compose([
            transforms.ToTensor(),
            transforms.Normalize(train_mean, train_std),
        ]),
    ),
    batch_size=args.test_batch_size,
    shuffle=False,
)
```

```
100.0%
100.0%
100.0%
100.0%
```

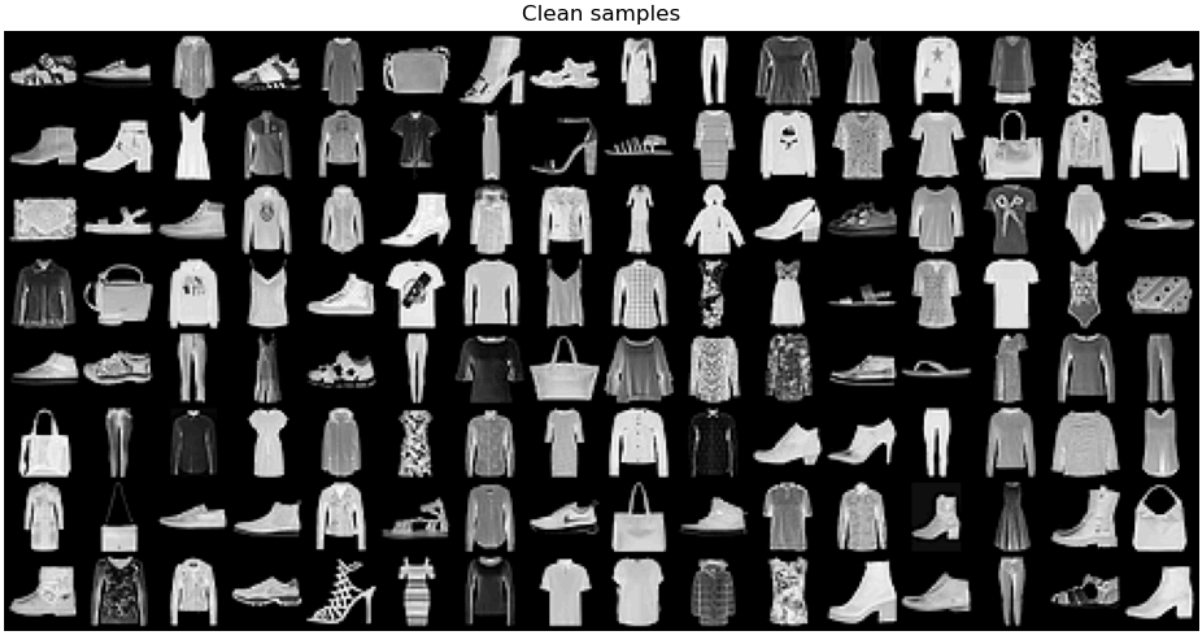
In [6]:
```
# Display a clean and noisy batch.
in_img = next(iter(train_loader))[0]
inputs = torch.stack([UnNormalize(train_mean, train_std)(in_img[i].clone())
noisy_inputs = AddWhiteNoise(noise_factor=args.noise_factor)(inputs.clone())

plt.imshow(transforms.ToPILImage()(make_grid(noisy_inputs, nrow=16)))
plt.title('Noisy samples')
plt.axis('off')
```

```
plt.show()

plt.imshow(transforms.ToPILImage()(make_grid(inputs, nrow=16)))
plt.title('Clean samples')
plt.axis('off')
plt.show()
```

Noisy samples



Clean samples



Task 1: Fully Connected Autoencoder

Implement the `Autoencoder` class below.

Encoder architecture:

- `Linear(784, 128)` + `ReLU`

- `Linear(128, 64)` + `ReLU`
- `Linear(64, 32)` + `ReLU`
- `Linear(32, 16)` + `ReLU`
- `Linear(16, 8)` + `ReLU`

Decoder architecture:

- `Linear(8, 8)` + `ReLU`
- `Linear(8, 32)` + `ReLU`
- `Linear(32, 64)` + `ReLU`
- `Linear(64, 128)` + `ReLU`
- `Linear(128, 784)` + `Tanh`

In [8]:
```
class Autoencoder(nn.Module):
    def __init__(self, input_shape=(28, 28)):
        super(Autoencoder, self).__init__()
        input_dim = int(np.prod(input_shape))

        encoder_layers = [
            nn.Linear(input_dim, 128),
            nn.ReLU(),
            nn.Linear(128, 64),
            nn.ReLU(),
            nn.Linear(64, 32),
            nn.ReLU(),
            nn.Linear(32, 16),
            nn.ReLU(),
            nn.Linear(16, 8),
            nn.ReLU()
        ]
        if not encoder_layers:
            raise NotImplementedError('TODO 1.1: implement the fully connect
        self.encoder = nn.Sequential(*encoder_layers)

        decoder_layers = [
            nn.Linear(8, 8),
            nn.ReLU(),
            nn.Linear(8, 32),
            nn.ReLU(),
            nn.Linear(32, 64),
            nn.ReLU(),
            nn.Linear(64, 128),
            nn.ReLU(),
            nn.Linear(128, input_dim),
            nn.Tanh()
        ]
        if not decoder_layers:
            raise NotImplementedError('TODO 1.2: implement the fully connect
        self.decoder = nn.Sequential(*decoder_layers)

    def forward(self, x):
        latent_code = self.encoder(x)
```

```
        reconstruction = self.decoder(latent_code)
        return reconstruction
```

In [9]:
```
# Run this after completing Task 1. The output shape must match the flattene
fc_shape_test = Autoencoder()
with torch.no_grad():
    dummy = torch.randn(4, 28 * 28)
    assert fc_shape_test(dummy).shape == dummy.shape
print('Autoencoder shape check passed.')
```

```
Autoencoder shape check passed.
```

Training and Evaluation Code

These functions are shared by all three tasks. You do not need to edit them.

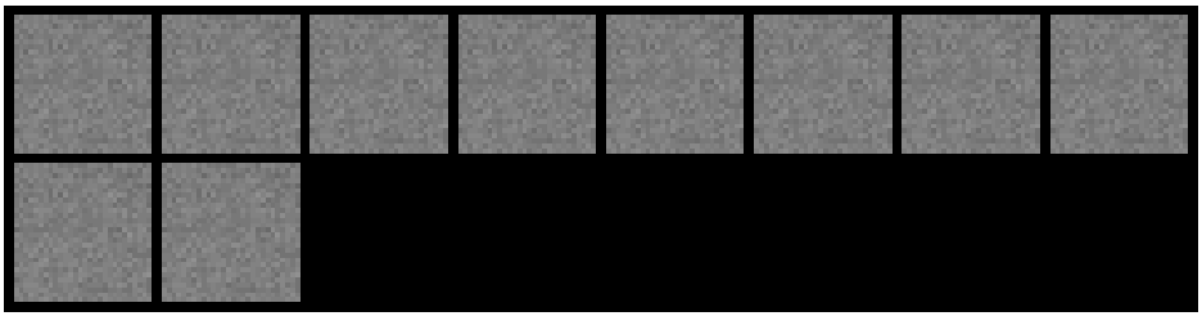
In [10]:
```
def initialize_training(model):
    model = model.to(args.device)
    optimizer = optim.Adam(model.parameters(), lr=args.lr, weight_decay=args
    criterion = nn.MSELoss()
    return model, optimizer, criterion


def prepare_batch(data, noisy=False):
    data = data.view(data.size(0), -1).to(args.device)
    if noisy:
        data_in = add_white_noise(data.clone(), factor=args.noise_factor)
        data_in.clamp_(-train_mean[0] / train_std[0], (1 - train_mean[0]) /
    else:
        data_in = data
    return data_in, data


def train_one_epoch(model, optimizer, criterion, epoch, noisy=False):
    model.train()
    running_loss = 0.0

    for batch_idx, (data, _) in enumerate(train_loader):
        data_in, target = prepare_batch(data, noisy=noisy)
        output = model(data_in)
        loss = criterion(output, target)

        optimizer.zero_grad()
        loss.backward()
        optimizer.step()

        running_loss += loss.item()

        if batch_idx % args.log_interval == 0:
            print(
                'Train Epoch: {} [{}/{} ({:.0f}%)]\tLoss: {:.6f}'.format(
                    epoch,
                    batch_idx * len(data),
                    len(train_loader.dataset),
```

```
                    100.0 * batch_idx / len(train_loader),
                    loss.item(),
                )
            )

    avg_loss = running_loss / len(train_loader)
    print('Train Epoch: {} Average loss: {:.6f}'.format(epoch, avg_loss))
    return avg_loss


def plot_reconstruction(data_in, output, target, noisy=False):
    if noisy:
        plt.imshow(transforms.ToPILImage()(make_grid(to_img(data_in.cpu()))))
        plt.title('Noisy input')
        plt.axis('off')
        plt.show()

    plt.imshow(transforms.ToPILImage()(make_grid(to_img(output.cpu()))))
    plt.title('Reconstruction')
    plt.axis('off')
    plt.show()

    plt.imshow(transforms.ToPILImage()(make_grid(to_img(target.cpu()))))
    plt.title('Target')
    plt.axis('off')
    plt.show()


def test(model, show_plot=False, noisy=False):
    model.eval()
    test_loss = 0.0
    criterion_sum = nn.MSELoss(reduction='sum')

    with torch.no_grad():
        for i, (data, _) in enumerate(test_loader):
            data_in, target = prepare_batch(data, noisy=noisy)
            output = model(data_in)
            test_loss += criterion_sum(output, target).item()

            if show_plot and i == 0:
                plot_reconstruction(data_in, output, target, noisy=noisy)

    test_loss /= len(test_loader.dataset)
    print('\nTest set: Average loss: {:.4f}\n'.format(test_loss))
    return test_loss


def run_experiment(model, save_path, noisy=False):
    model, optimizer, criterion = initialize_training(model)
    best_loss = float('inf')

    test(model, show_plot=True, noisy=noisy)
    for epoch in range(1, args.epochs + 1):
        train_one_epoch(model, optimizer, criterion, epoch, noisy=noisy)
        test_loss = test(model, show_plot=(epoch % 10 == 0), noisy=noisy)
```

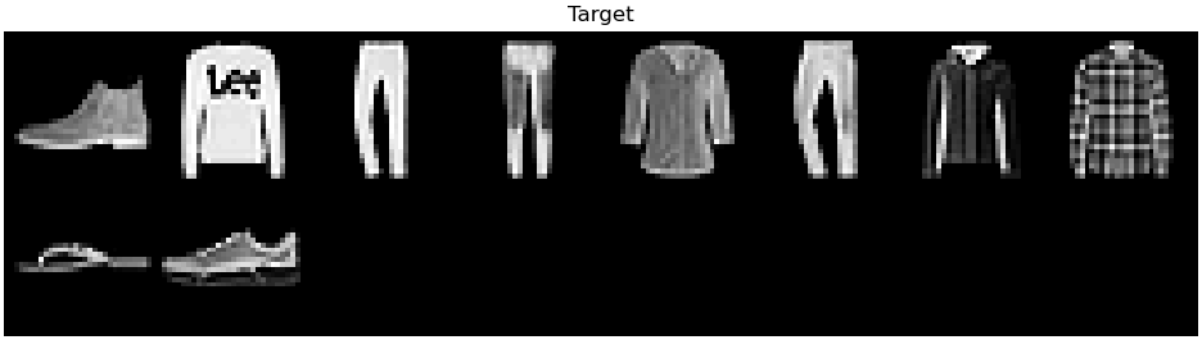
```
        if test_loss < best_loss:
            best_loss = test_loss
            torch.save(model.state_dict(), save_path)

    print('The best (lowest) achieved loss: {:.3f}\n'.format(best_loss))
    return best_loss
```

In [11]:
```
# Task 1 training: fully connected autoencoder.
args.is_noisy = False
best_fc_loss = run_experiment(
    Autoencoder(),
    save_path='best_fc_autoencoder.pth',
    noisy=args.is_noisy,
)
```

Reconstruction



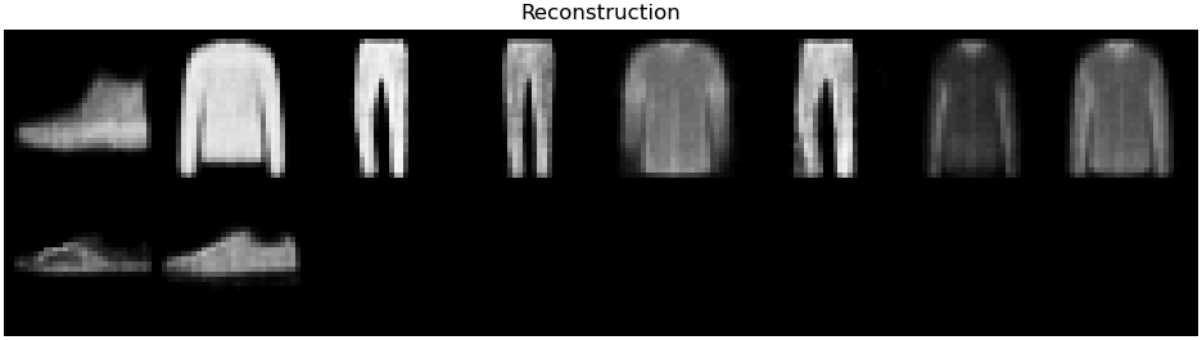
Target



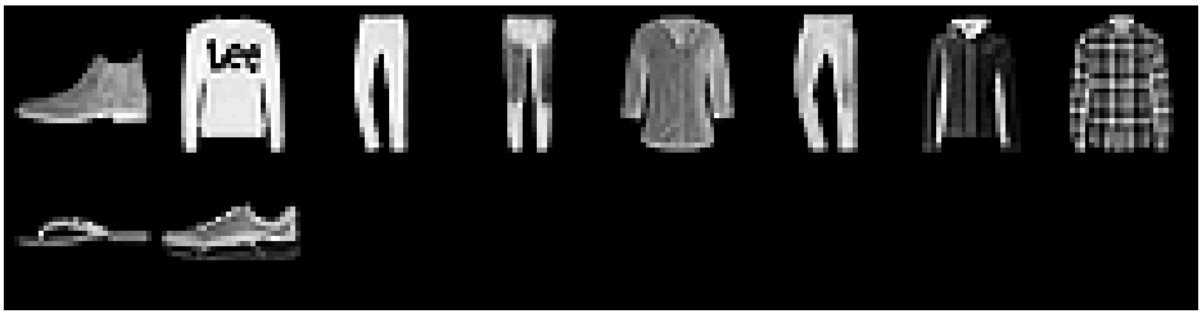
```
Test set: Average loss: 535.9782

Train Epoch: 1 [0/60000 (0%)]   Loss: 0.686385
Train Epoch: 1 [12800/60000 (21%)]      Loss: 0.273120
Train Epoch: 1 [25600/60000 (43%)]      Loss: 0.167884
Train Epoch: 1 [38400/60000 (64%)]      Loss: 0.169244
Train Epoch: 1 [51200/60000 (85%)]      Loss: 0.158591
Train Epoch: 1 Average loss: 0.215010

Test set: Average loss: 114.1775

Train Epoch: 2 [0/60000 (0%)]   Loss: 0.137671
Train Epoch: 2 [12800/60000 (21%)]      Loss: 0.152422
Train Epoch: 2 [25600/60000 (43%)]      Loss: 0.130134
Train Epoch: 2 [38400/60000 (64%)]      Loss: 0.114292
Train Epoch: 2 [51200/60000 (85%)]      Loss: 0.105804
Train Epoch: 2 Average loss: 0.128553

Test set: Average loss: 90.8615

Train Epoch: 3 [0/60000 (0%)]   Loss: 0.121524
Train Epoch: 3 [12800/60000 (21%)]      Loss: 0.118456
Train Epoch: 3 [25600/60000 (43%)]      Loss: 0.099250
Train Epoch: 3 [38400/60000 (64%)]      Loss: 0.111304
Train Epoch: 3 [51200/60000 (85%)]      Loss: 0.100251
Train Epoch: 3 Average loss: 0.106680

Test set: Average loss: 76.3064

Train Epoch: 4 [0/60000 (0%)]   Loss: 0.103554
Train Epoch: 4 [12800/60000 (21%)]      Loss: 0.093935
Train Epoch: 4 [25600/60000 (43%)]      Loss: 0.097772
Train Epoch: 4 [38400/60000 (64%)]      Loss: 0.089139
Train Epoch: 4 [51200/60000 (85%)]      Loss: 0.096051
Train Epoch: 4 Average loss: 0.094715

Test set: Average loss: 73.1442

Train Epoch: 5 [0/60000 (0%)]   Loss: 0.093151
Train Epoch: 5 [12800/60000 (21%)]      Loss: 0.100141
Train Epoch: 5 [25600/60000 (43%)]      Loss: 0.083341
Train Epoch: 5 [38400/60000 (64%)]      Loss: 0.100641
Train Epoch: 5 [51200/60000 (85%)]      Loss: 0.096413
Train Epoch: 5 Average loss: 0.091045

Test set: Average loss: 70.6957

Train Epoch: 6 [0/60000 (0%)]   Loss: 0.086920
Train Epoch: 6 [12800/60000 (21%)]      Loss: 0.091931
Train Epoch: 6 [25600/60000 (43%)]      Loss: 0.082438
Train Epoch: 6 [38400/60000 (64%)]      Loss: 0.088127
Train Epoch: 6 [51200/60000 (85%)]      Loss: 0.090505
Train Epoch: 6 Average loss: 0.088707

Test set: Average loss: 69.3981
```

```
Train Epoch: 7 [0/60000 (0%)]   Loss: 0.089447
Train Epoch: 7 [12800/60000 (21%)]      Loss: 0.087402
Train Epoch: 7 [25600/60000 (43%)]      Loss: 0.083732
Train Epoch: 7 [38400/60000 (64%)]      Loss: 0.086954
Train Epoch: 7 [51200/60000 (85%)]      Loss: 0.080389
Train Epoch: 7 Average loss: 0.086969

Test set: Average loss: 67.7925

Train Epoch: 8 [0/60000 (0%)]   Loss: 0.083488
Train Epoch: 8 [12800/60000 (21%)]      Loss: 0.090174
Train Epoch: 8 [25600/60000 (43%)]      Loss: 0.082713
Train Epoch: 8 [38400/60000 (64%)]      Loss: 0.091027
Train Epoch: 8 [51200/60000 (85%)]      Loss: 0.092466
Train Epoch: 8 Average loss: 0.085740

Test set: Average loss: 67.4481

Train Epoch: 9 [0/60000 (0%)]   Loss: 0.089544
Train Epoch: 9 [12800/60000 (21%)]      Loss: 0.087410
Train Epoch: 9 [25600/60000 (43%)]      Loss: 0.076994
Train Epoch: 9 [38400/60000 (64%)]      Loss: 0.089280
Train Epoch: 9 [51200/60000 (85%)]      Loss: 0.083110
Train Epoch: 9 Average loss: 0.084908

Test set: Average loss: 66.5952

Train Epoch: 10 [0/60000 (0%)]  Loss: 0.082615
Train Epoch: 10 [12800/60000 (21%)]     Loss: 0.072086
Train Epoch: 10 [25600/60000 (43%)]     Loss: 0.086174
Train Epoch: 10 [38400/60000 (64%)]     Loss: 0.076565
Train Epoch: 10 [51200/60000 (85%)]     Loss: 0.080159
Train Epoch: 10 Average loss: 0.084011
```

Reconstruction



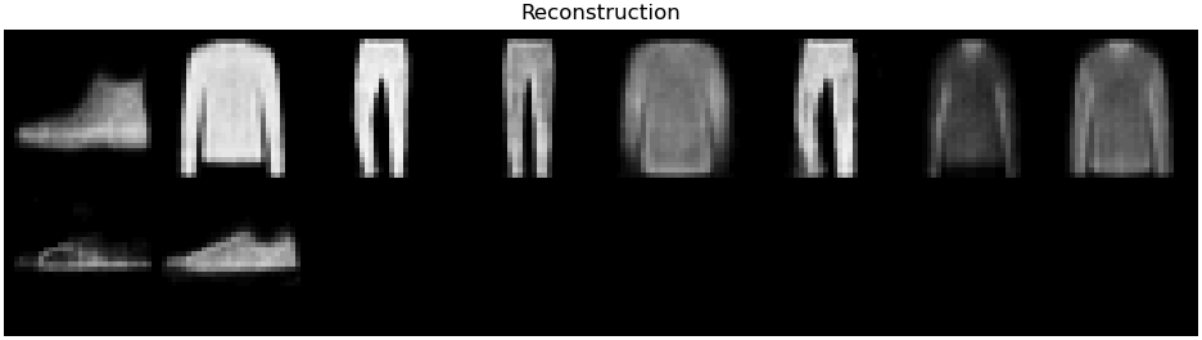
Target



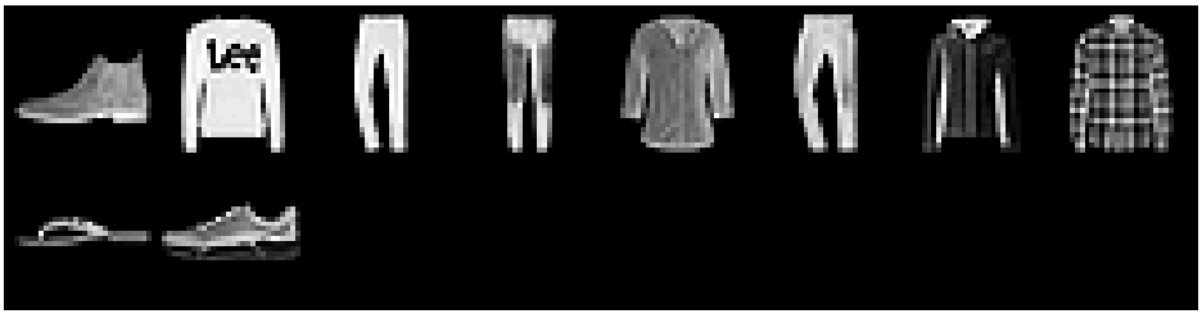
```
Test set: Average loss: 65.9858

Train Epoch: 11 [0/60000 (0%)]  Loss: 0.082488
Train Epoch: 11 [12800/60000 (21%)]     Loss: 0.085228
Train Epoch: 11 [25600/60000 (43%)]     Loss: 0.082092
Train Epoch: 11 [38400/60000 (64%)]     Loss: 0.079595
Train Epoch: 11 [51200/60000 (85%)]     Loss: 0.083854
Train Epoch: 11 Average loss: 0.083405

Test set: Average loss: 65.4256

Train Epoch: 12 [0/60000 (0%)]  Loss: 0.083901
Train Epoch: 12 [12800/60000 (21%)]     Loss: 0.085419
Train Epoch: 12 [25600/60000 (43%)]     Loss: 0.078721
Train Epoch: 12 [38400/60000 (64%)]     Loss: 0.076620
Train Epoch: 12 [51200/60000 (85%)]     Loss: 0.083110
Train Epoch: 12 Average loss: 0.082772

Test set: Average loss: 65.1708

Train Epoch: 13 [0/60000 (0%)]  Loss: 0.085366
Train Epoch: 13 [12800/60000 (21%)]     Loss: 0.075168
Train Epoch: 13 [25600/60000 (43%)]     Loss: 0.077800
Train Epoch: 13 [38400/60000 (64%)]     Loss: 0.086533
Train Epoch: 13 [51200/60000 (85%)]     Loss: 0.084516
Train Epoch: 13 Average loss: 0.082185

Test set: Average loss: 64.6509

Train Epoch: 14 [0/60000 (0%)]  Loss: 0.080916
Train Epoch: 14 [12800/60000 (21%)]     Loss: 0.086887
Train Epoch: 14 [25600/60000 (43%)]     Loss: 0.088726
Train Epoch: 14 [38400/60000 (64%)]     Loss: 0.079835
Train Epoch: 14 [51200/60000 (85%)]     Loss: 0.084808
Train Epoch: 14 Average loss: 0.081731

Test set: Average loss: 64.2172

Train Epoch: 15 [0/60000 (0%)]  Loss: 0.074637
Train Epoch: 15 [12800/60000 (21%)]     Loss: 0.081229
Train Epoch: 15 [25600/60000 (43%)]     Loss: 0.077195
Train Epoch: 15 [38400/60000 (64%)]     Loss: 0.082128
Train Epoch: 15 [51200/60000 (85%)]     Loss: 0.087292
Train Epoch: 15 Average loss: 0.081266

Test set: Average loss: 64.0788

Train Epoch: 16 [0/60000 (0%)]  Loss: 0.078828
Train Epoch: 16 [12800/60000 (21%)]     Loss: 0.087558
Train Epoch: 16 [25600/60000 (43%)]     Loss: 0.089388
Train Epoch: 16 [38400/60000 (64%)]     Loss: 0.083379
Train Epoch: 16 [51200/60000 (85%)]     Loss: 0.086723
Train Epoch: 16 Average loss: 0.080929

Test set: Average loss: 63.6301
```

```
Train Epoch: 17 [0/60000 (0%)]  Loss: 0.087995
Train Epoch: 17 [12800/60000 (21%)]     Loss: 0.081877
Train Epoch: 17 [25600/60000 (43%)]     Loss: 0.082034
Train Epoch: 17 [38400/60000 (64%)]     Loss: 0.082002
Train Epoch: 17 [51200/60000 (85%)]     Loss: 0.083260
Train Epoch: 17 Average loss: 0.080541

Test set: Average loss: 63.5398

Train Epoch: 18 [0/60000 (0%)]  Loss: 0.080398
Train Epoch: 18 [12800/60000 (21%)]     Loss: 0.075995
Train Epoch: 18 [25600/60000 (43%)]     Loss: 0.083494
Train Epoch: 18 [38400/60000 (64%)]     Loss: 0.083286
Train Epoch: 18 [51200/60000 (85%)]     Loss: 0.079252
Train Epoch: 18 Average loss: 0.080089

Test set: Average loss: 63.1554

Train Epoch: 19 [0/60000 (0%)]  Loss: 0.082208
Train Epoch: 19 [12800/60000 (21%)]     Loss: 0.076089
Train Epoch: 19 [25600/60000 (43%)]     Loss: 0.077209
Train Epoch: 19 [38400/60000 (64%)]     Loss: 0.082021
Train Epoch: 19 [51200/60000 (85%)]     Loss: 0.079392
Train Epoch: 19 Average loss: 0.079854

Test set: Average loss: 62.8084

Train Epoch: 20 [0/60000 (0%)]  Loss: 0.076311
Train Epoch: 20 [12800/60000 (21%)]     Loss: 0.081581
Train Epoch: 20 [25600/60000 (43%)]     Loss: 0.075558
Train Epoch: 20 [38400/60000 (64%)]     Loss: 0.075061
Train Epoch: 20 [51200/60000 (85%)]     Loss: 0.079752
Train Epoch: 20 Average loss: 0.079608
```

Reconstruction



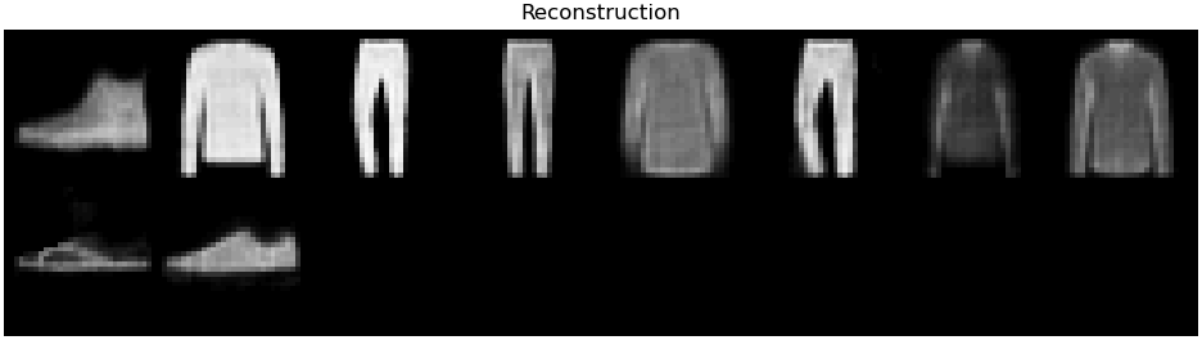
Target



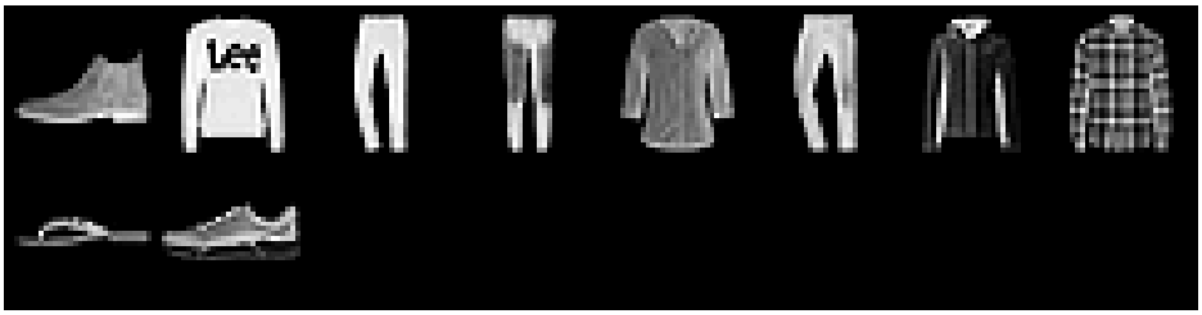
```
Test set: Average loss: 62.9091

Train Epoch: 21 [0/60000 (0%)]  Loss: 0.074797
Train Epoch: 21 [12800/60000 (21%)]     Loss: 0.075360
Train Epoch: 21 [25600/60000 (43%)]     Loss: 0.080101
Train Epoch: 21 [38400/60000 (64%)]     Loss: 0.074794
Train Epoch: 21 [51200/60000 (85%)]     Loss: 0.085834
Train Epoch: 21 Average loss: 0.079262

Test set: Average loss: 62.4690

Train Epoch: 22 [0/60000 (0%)]  Loss: 0.080885
Train Epoch: 22 [12800/60000 (21%)]     Loss: 0.082110
Train Epoch: 22 [25600/60000 (43%)]     Loss: 0.078992
Train Epoch: 22 [38400/60000 (64%)]     Loss: 0.080199
Train Epoch: 22 [51200/60000 (85%)]     Loss: 0.079424
Train Epoch: 22 Average loss: 0.079029

Test set: Average loss: 62.2984

Train Epoch: 23 [0/60000 (0%)]  Loss: 0.083372
Train Epoch: 23 [12800/60000 (21%)]     Loss: 0.074692
Train Epoch: 23 [25600/60000 (43%)]     Loss: 0.079500
Train Epoch: 23 [38400/60000 (64%)]     Loss: 0.076736
Train Epoch: 23 [51200/60000 (85%)]     Loss: 0.081062
Train Epoch: 23 Average loss: 0.078722

Test set: Average loss: 62.5785

Train Epoch: 24 [0/60000 (0%)]  Loss: 0.076987
Train Epoch: 24 [12800/60000 (21%)]     Loss: 0.083476
Train Epoch: 24 [25600/60000 (43%)]     Loss: 0.074409
Train Epoch: 24 [38400/60000 (64%)]     Loss: 0.078993
Train Epoch: 24 [51200/60000 (85%)]     Loss: 0.079910
Train Epoch: 24 Average loss: 0.078612

Test set: Average loss: 62.2359

Train Epoch: 25 [0/60000 (0%)]  Loss: 0.075229
Train Epoch: 25 [12800/60000 (21%)]     Loss: 0.079885
Train Epoch: 25 [25600/60000 (43%)]     Loss: 0.077187
Train Epoch: 25 [38400/60000 (64%)]     Loss: 0.075169
Train Epoch: 25 [51200/60000 (85%)]     Loss: 0.076300
Train Epoch: 25 Average loss: 0.078359

Test set: Average loss: 62.0985

Train Epoch: 26 [0/60000 (0%)]  Loss: 0.082406
Train Epoch: 26 [12800/60000 (21%)]     Loss: 0.069869
Train Epoch: 26 [25600/60000 (43%)]     Loss: 0.079270
Train Epoch: 26 [38400/60000 (64%)]     Loss: 0.073007
Train Epoch: 26 [51200/60000 (85%)]     Loss: 0.067462
Train Epoch: 26 Average loss: 0.078144

Test set: Average loss: 61.7947
```

```
Train Epoch: 27 [0/60000 (0%)]  Loss: 0.077654
Train Epoch: 27 [12800/60000 (21%)]     Loss: 0.078764
Train Epoch: 27 [25600/60000 (43%)]     Loss: 0.073906
Train Epoch: 27 [38400/60000 (64%)]     Loss: 0.080656
Train Epoch: 27 [51200/60000 (85%)]     Loss: 0.083414
Train Epoch: 27 Average loss: 0.077962

Test set: Average loss: 61.5122

Train Epoch: 28 [0/60000 (0%)]  Loss: 0.079506
Train Epoch: 28 [12800/60000 (21%)]     Loss: 0.080424
Train Epoch: 28 [25600/60000 (43%)]     Loss: 0.079911
Train Epoch: 28 [38400/60000 (64%)]     Loss: 0.076675
Train Epoch: 28 [51200/60000 (85%)]     Loss: 0.069951
Train Epoch: 28 Average loss: 0.077816

Test set: Average loss: 61.8985

Train Epoch: 29 [0/60000 (0%)]  Loss: 0.075301
Train Epoch: 29 [12800/60000 (21%)]     Loss: 0.082065
Train Epoch: 29 [25600/60000 (43%)]     Loss: 0.085745
Train Epoch: 29 [38400/60000 (64%)]     Loss: 0.076163
Train Epoch: 29 [51200/60000 (85%)]     Loss: 0.078664
Train Epoch: 29 Average loss: 0.077672

Test set: Average loss: 61.4040

Train Epoch: 30 [0/60000 (0%)]  Loss: 0.078536
Train Epoch: 30 [12800/60000 (21%)]     Loss: 0.076590
Train Epoch: 30 [25600/60000 (43%)]     Loss: 0.077624
Train Epoch: 30 [38400/60000 (64%)]     Loss: 0.079587
Train Epoch: 30 [51200/60000 (85%)]     Loss: 0.073271
Train Epoch: 30 Average loss: 0.077487
```

Reconstruction



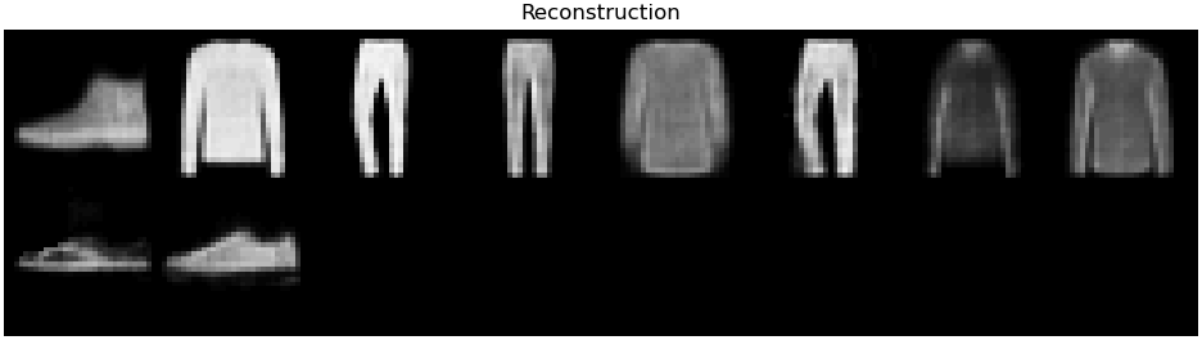
Target



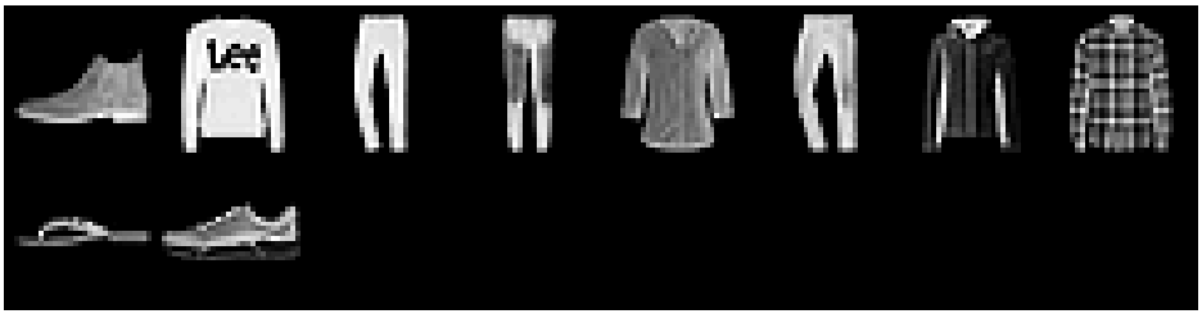
```
Test set: Average loss: 61.2044

Train Epoch: 31 [0/60000 (0%)]  Loss: 0.084164
Train Epoch: 31 [12800/60000 (21%)]     Loss: 0.081247
Train Epoch: 31 [25600/60000 (43%)]     Loss: 0.075194
Train Epoch: 31 [38400/60000 (64%)]     Loss: 0.072455
Train Epoch: 31 [51200/60000 (85%)]     Loss: 0.076890
Train Epoch: 31 Average loss: 0.077318

Test set: Average loss: 61.2403

Train Epoch: 32 [0/60000 (0%)]  Loss: 0.072726
Train Epoch: 32 [12800/60000 (21%)]     Loss: 0.070355
Train Epoch: 32 [25600/60000 (43%)]     Loss: 0.075949
Train Epoch: 32 [38400/60000 (64%)]     Loss: 0.073315
Train Epoch: 32 [51200/60000 (85%)]     Loss: 0.073312
Train Epoch: 32 Average loss: 0.077144

Test set: Average loss: 61.4094

Train Epoch: 33 [0/60000 (0%)]  Loss: 0.079001
Train Epoch: 33 [12800/60000 (21%)]     Loss: 0.079905
Train Epoch: 33 [25600/60000 (43%)]     Loss: 0.080007
Train Epoch: 33 [38400/60000 (64%)]     Loss: 0.081161
Train Epoch: 33 [51200/60000 (85%)]     Loss: 0.069343
Train Epoch: 33 Average loss: 0.077078

Test set: Average loss: 60.8209

Train Epoch: 34 [0/60000 (0%)]  Loss: 0.076787
Train Epoch: 34 [12800/60000 (21%)]     Loss: 0.088540
Train Epoch: 34 [25600/60000 (43%)]     Loss: 0.082845
Train Epoch: 34 [38400/60000 (64%)]     Loss: 0.077471
Train Epoch: 34 [51200/60000 (85%)]     Loss: 0.081957
Train Epoch: 34 Average loss: 0.076925

Test set: Average loss: 61.2036

Train Epoch: 35 [0/60000 (0%)]  Loss: 0.079037
Train Epoch: 35 [12800/60000 (21%)]     Loss: 0.074244
Train Epoch: 35 [25600/60000 (43%)]     Loss: 0.081408
Train Epoch: 35 [38400/60000 (64%)]     Loss: 0.076666
Train Epoch: 35 [51200/60000 (85%)]     Loss: 0.074086
Train Epoch: 35 Average loss: 0.076745

Test set: Average loss: 60.9947

Train Epoch: 36 [0/60000 (0%)]  Loss: 0.083585
Train Epoch: 36 [12800/60000 (21%)]     Loss: 0.075060
Train Epoch: 36 [25600/60000 (43%)]     Loss: 0.076440
Train Epoch: 36 [38400/60000 (64%)]     Loss: 0.085559
Train Epoch: 36 [51200/60000 (85%)]     Loss: 0.072433
Train Epoch: 36 Average loss: 0.076627

Test set: Average loss: 60.7698
```

```
Train Epoch: 37 [0/60000 (0%)]  Loss: 0.081009
Train Epoch: 37 [12800/60000 (21%)]     Loss: 0.080894
Train Epoch: 37 [25600/60000 (43%)]     Loss: 0.069941
Train Epoch: 37 [38400/60000 (64%)]     Loss: 0.071441
Train Epoch: 37 [51200/60000 (85%)]     Loss: 0.080008
Train Epoch: 37 Average loss: 0.076524

Test set: Average loss: 61.2202

Train Epoch: 38 [0/60000 (0%)]  Loss: 0.075012
Train Epoch: 38 [12800/60000 (21%)]     Loss: 0.071859
Train Epoch: 38 [25600/60000 (43%)]     Loss: 0.071062
Train Epoch: 38 [38400/60000 (64%)]     Loss: 0.069534
Train Epoch: 38 [51200/60000 (85%)]     Loss: 0.076985
Train Epoch: 38 Average loss: 0.076502

Test set: Average loss: 61.2077

Train Epoch: 39 [0/60000 (0%)]  Loss: 0.079553
Train Epoch: 39 [12800/60000 (21%)]     Loss: 0.070103
Train Epoch: 39 [25600/60000 (43%)]     Loss: 0.081536
Train Epoch: 39 [38400/60000 (64%)]     Loss: 0.076367
Train Epoch: 39 [51200/60000 (85%)]     Loss: 0.079660
Train Epoch: 39 Average loss: 0.076361

Test set: Average loss: 60.8070

Train Epoch: 40 [0/60000 (0%)]  Loss: 0.078317
Train Epoch: 40 [12800/60000 (21%)]     Loss: 0.073854
Train Epoch: 40 [25600/60000 (43%)]     Loss: 0.071159
Train Epoch: 40 [38400/60000 (64%)]     Loss: 0.076664
Train Epoch: 40 [51200/60000 (85%)]     Loss: 0.081968
Train Epoch: 40 Average loss: 0.076184
```

Reconstruction



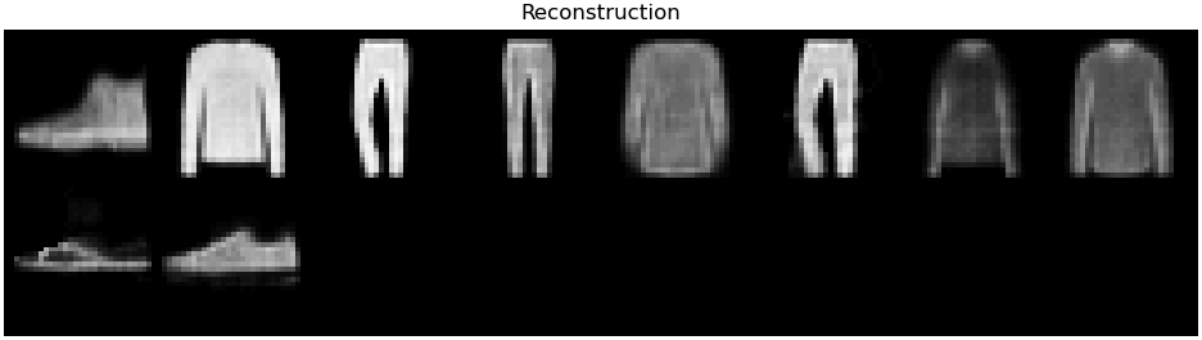
Target



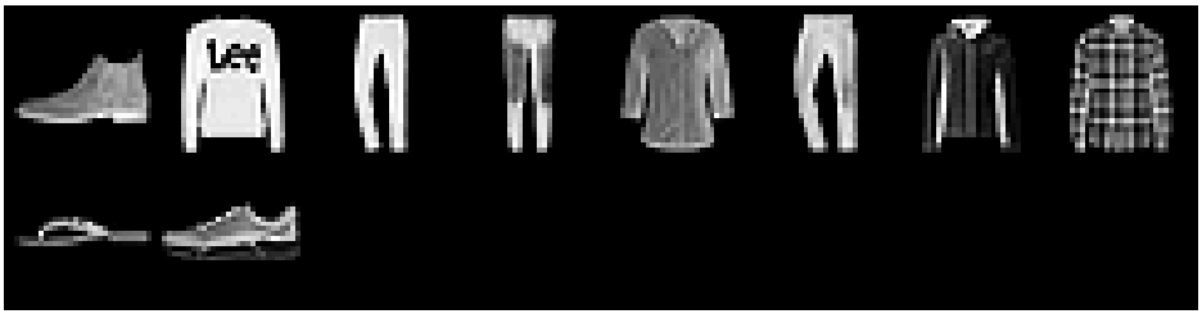
```
Test set: Average loss: 60.4193

Train Epoch: 41 [0/60000 (0%)]  Loss: 0.078517
Train Epoch: 41 [12800/60000 (21%)]     Loss: 0.073714
Train Epoch: 41 [25600/60000 (43%)]     Loss: 0.073572
Train Epoch: 41 [38400/60000 (64%)]     Loss: 0.079123
Train Epoch: 41 [51200/60000 (85%)]     Loss: 0.074492
Train Epoch: 41 Average loss: 0.076094

Test set: Average loss: 60.4443

Train Epoch: 42 [0/60000 (0%)]  Loss: 0.073679
Train Epoch: 42 [12800/60000 (21%)]     Loss: 0.072803
Train Epoch: 42 [25600/60000 (43%)]     Loss: 0.073667
Train Epoch: 42 [38400/60000 (64%)]     Loss: 0.076583
Train Epoch: 42 [51200/60000 (85%)]     Loss: 0.077789
Train Epoch: 42 Average loss: 0.076037

Test set: Average loss: 60.4705

Train Epoch: 43 [0/60000 (0%)]  Loss: 0.077419
Train Epoch: 43 [12800/60000 (21%)]     Loss: 0.075163
Train Epoch: 43 [25600/60000 (43%)]     Loss: 0.071838
Train Epoch: 43 [38400/60000 (64%)]     Loss: 0.082948
Train Epoch: 43 [51200/60000 (85%)]     Loss: 0.077800
Train Epoch: 43 Average loss: 0.075966

Test set: Average loss: 60.3222

Train Epoch: 44 [0/60000 (0%)]  Loss: 0.082242
Train Epoch: 44 [12800/60000 (21%)]     Loss: 0.076267
Train Epoch: 44 [25600/60000 (43%)]     Loss: 0.072965
Train Epoch: 44 [38400/60000 (64%)]     Loss: 0.075671
Train Epoch: 44 [51200/60000 (85%)]     Loss: 0.077671
Train Epoch: 44 Average loss: 0.075816

Test set: Average loss: 60.2065

Train Epoch: 45 [0/60000 (0%)]  Loss: 0.081520
Train Epoch: 45 [12800/60000 (21%)]     Loss: 0.074756
Train Epoch: 45 [25600/60000 (43%)]     Loss: 0.075784
Train Epoch: 45 [38400/60000 (64%)]     Loss: 0.075365
Train Epoch: 45 [51200/60000 (85%)]     Loss: 0.072108
Train Epoch: 45 Average loss: 0.075650

Test set: Average loss: 60.0606

Train Epoch: 46 [0/60000 (0%)]  Loss: 0.075871
Train Epoch: 46 [12800/60000 (21%)]     Loss: 0.077699
Train Epoch: 46 [25600/60000 (43%)]     Loss: 0.078151
Train Epoch: 46 [38400/60000 (64%)]     Loss: 0.071699
Train Epoch: 46 [51200/60000 (85%)]     Loss: 0.085244
Train Epoch: 46 Average loss: 0.075557

Test set: Average loss: 60.0697
```

```
Train Epoch: 47 [0/60000 (0%)]  Loss: 0.075399
Train Epoch: 47 [12800/60000 (21%)]     Loss: 0.076898
Train Epoch: 47 [25600/60000 (43%)]     Loss: 0.077482
Train Epoch: 47 [38400/60000 (64%)]     Loss: 0.073673
Train Epoch: 47 [51200/60000 (85%)]     Loss: 0.073548
Train Epoch: 47 Average loss: 0.075469

Test set: Average loss: 60.2513

Train Epoch: 48 [0/60000 (0%)]  Loss: 0.077056
Train Epoch: 48 [12800/60000 (21%)]     Loss: 0.069551
Train Epoch: 48 [25600/60000 (43%)]     Loss: 0.074976
Train Epoch: 48 [38400/60000 (64%)]     Loss: 0.073865
Train Epoch: 48 [51200/60000 (85%)]     Loss: 0.080042
Train Epoch: 48 Average loss: 0.075449

Test set: Average loss: 60.3598

Train Epoch: 49 [0/60000 (0%)]  Loss: 0.084138
Train Epoch: 49 [12800/60000 (21%)]     Loss: 0.079942
Train Epoch: 49 [25600/60000 (43%)]     Loss: 0.074277
Train Epoch: 49 [38400/60000 (64%)]     Loss: 0.078494
Train Epoch: 49 [51200/60000 (85%)]     Loss: 0.068168
Train Epoch: 49 Average loss: 0.075326

Test set: Average loss: 60.3293

Train Epoch: 50 [0/60000 (0%)]  Loss: 0.074330
Train Epoch: 50 [12800/60000 (21%)]     Loss: 0.080847
Train Epoch: 50 [25600/60000 (43%)]     Loss: 0.064173
Train Epoch: 50 [38400/60000 (64%)]     Loss: 0.074595
Train Epoch: 50 [51200/60000 (85%)]     Loss: 0.074330
Train Epoch: 50 Average loss: 0.075269
```

Reconstruction



Target



```
Test set: Average loss: 59.9906

The best (lowest) achieved loss: 59.991
```

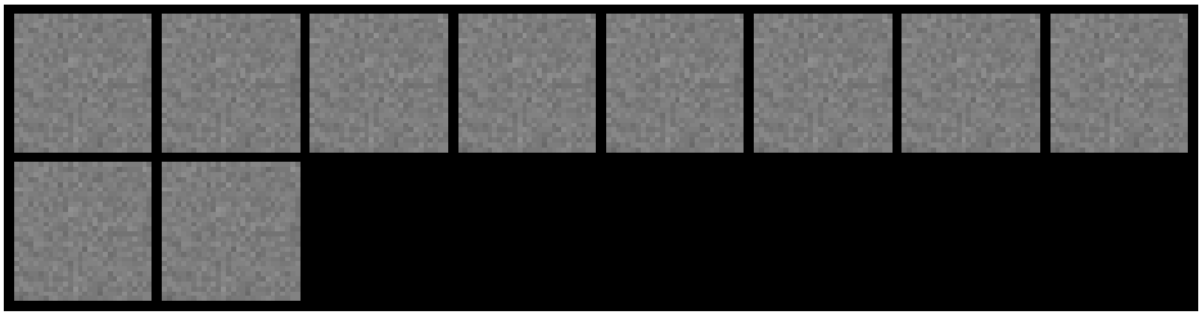
Task 2: Denoising Autoencoder

Use the same `Autoencoder` architecture from Task 1. This time, train the model with noisy inputs, but compute the reconstruction loss against the original clean image.

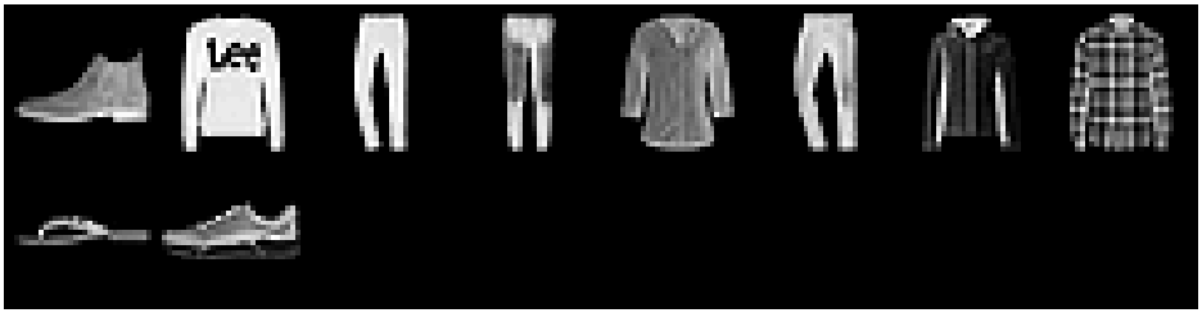
In [12]:
```
# Task 2 training: denoising autoencoder.
args.is_noisy = True
best_fc_denoising_loss = run_experiment(
    Autoencoder(),
    save_path='best_fc_denoising_autoencoder.pth',
    noisy=args.is_noisy,
)
```

Noisy input

Reconstruction



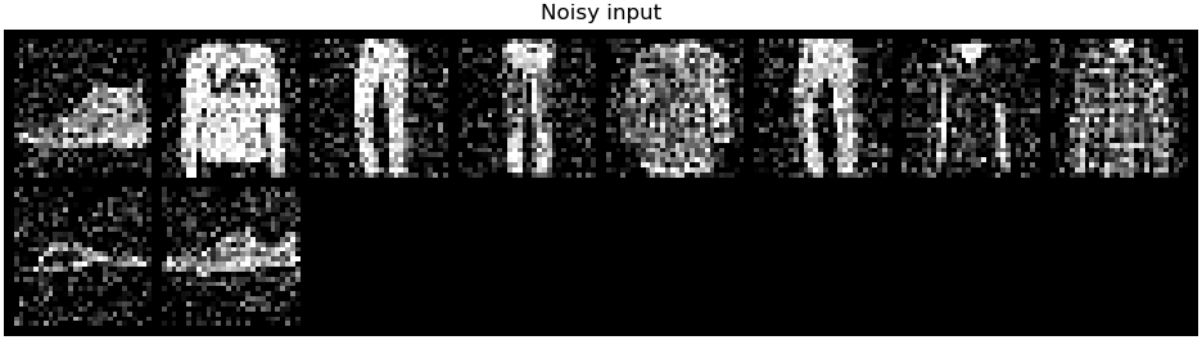
Target



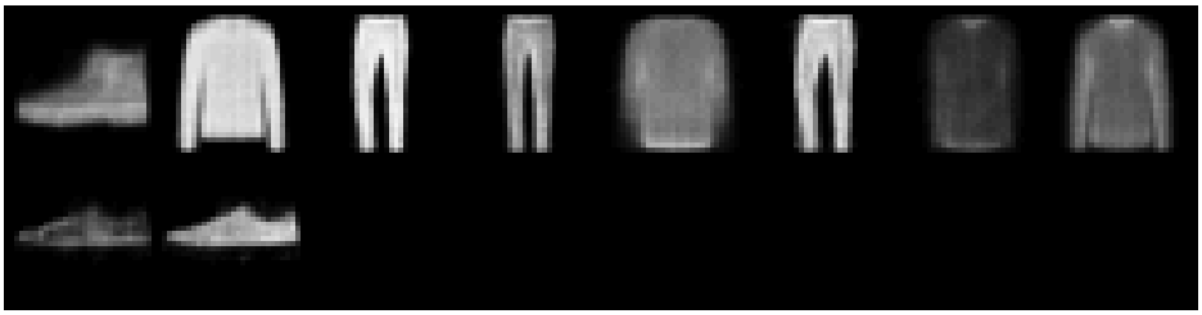
```
Test set: Average loss: 534.5256

Train Epoch: 1 [0/60000 (0%)]   Loss: 0.676098
Train Epoch: 1 [12800/60000 (21%)]      Loss: 0.235327
Train Epoch: 1 [25600/60000 (43%)]      Loss: 0.166709
Train Epoch: 1 [38400/60000 (64%)]      Loss: 0.146526
Train Epoch: 1 [51200/60000 (85%)]      Loss: 0.142850
Train Epoch: 1 Average loss: 0.206297

Test set: Average loss: 113.9774

Train Epoch: 2 [0/60000 (0%)]   Loss: 0.143529
Train Epoch: 2 [12800/60000 (21%)]      Loss: 0.129860
Train Epoch: 2 [25600/60000 (43%)]      Loss: 0.125377
Train Epoch: 2 [38400/60000 (64%)]      Loss: 0.118073
Train Epoch: 2 [51200/60000 (85%)]      Loss: 0.125623
Train Epoch: 2 Average loss: 0.129466

Test set: Average loss: 92.4380

Train Epoch: 3 [0/60000 (0%)]   Loss: 0.111914
Train Epoch: 3 [12800/60000 (21%)]      Loss: 0.103188
Train Epoch: 3 [25600/60000 (43%)]      Loss: 0.121973
Train Epoch: 3 [38400/60000 (64%)]      Loss: 0.105286
Train Epoch: 3 [51200/60000 (85%)]      Loss: 0.110865
Train Epoch: 3 Average loss: 0.113227

Test set: Average loss: 85.2863

Train Epoch: 4 [0/60000 (0%)]   Loss: 0.113306
Train Epoch: 4 [12800/60000 (21%)]      Loss: 0.105940
Train Epoch: 4 [25600/60000 (43%)]      Loss: 0.105860
Train Epoch: 4 [38400/60000 (64%)]      Loss: 0.108886
Train Epoch: 4 [51200/60000 (85%)]      Loss: 0.109546
Train Epoch: 4 Average loss: 0.106860

Test set: Average loss: 82.2763

Train Epoch: 5 [0/60000 (0%)]   Loss: 0.105537
Train Epoch: 5 [12800/60000 (21%)]      Loss: 0.105006
Train Epoch: 5 [25600/60000 (43%)]      Loss: 0.101711
Train Epoch: 5 [38400/60000 (64%)]      Loss: 0.101902
Train Epoch: 5 [51200/60000 (85%)]      Loss: 0.107149
Train Epoch: 5 Average loss: 0.103443

Test set: Average loss: 80.5340

Train Epoch: 6 [0/60000 (0%)]   Loss: 0.100311
Train Epoch: 6 [12800/60000 (21%)]      Loss: 0.099764
Train Epoch: 6 [25600/60000 (43%)]      Loss: 0.104105
Train Epoch: 6 [38400/60000 (64%)]      Loss: 0.106342
Train Epoch: 6 [51200/60000 (85%)]      Loss: 0.101224
Train Epoch: 6 Average loss: 0.101286

Test set: Average loss: 78.6633
```

```
Train Epoch: 7 [0/60000 (0%)]   Loss: 0.103634
Train Epoch: 7 [12800/60000 (21%)]      Loss: 0.094386
Train Epoch: 7 [25600/60000 (43%)]      Loss: 0.110940
Train Epoch: 7 [38400/60000 (64%)]      Loss: 0.099714
Train Epoch: 7 [51200/60000 (85%)]      Loss: 0.098981
Train Epoch: 7 Average loss: 0.099637

Test set: Average loss: 78.0778

Train Epoch: 8 [0/60000 (0%)]   Loss: 0.099542
Train Epoch: 8 [12800/60000 (21%)]      Loss: 0.092695
Train Epoch: 8 [25600/60000 (43%)]      Loss: 0.102239
Train Epoch: 8 [38400/60000 (64%)]      Loss: 0.092392
Train Epoch: 8 [51200/60000 (85%)]      Loss: 0.098911
Train Epoch: 8 Average loss: 0.098035

Test set: Average loss: 76.7471

Train Epoch: 9 [0/60000 (0%)]   Loss: 0.101956
Train Epoch: 9 [12800/60000 (21%)]      Loss: 0.094685
Train Epoch: 9 [25600/60000 (43%)]      Loss: 0.105490
Train Epoch: 9 [38400/60000 (64%)]      Loss: 0.101010
Train Epoch: 9 [51200/60000 (85%)]      Loss: 0.099146
Train Epoch: 9 Average loss: 0.096809

Test set: Average loss: 75.7819

Train Epoch: 10 [0/60000 (0%)]  Loss: 0.102850
Train Epoch: 10 [12800/60000 (21%)]     Loss: 0.095458
Train Epoch: 10 [25600/60000 (43%)]     Loss: 0.099606
Train Epoch: 10 [38400/60000 (64%)]     Loss: 0.092644
Train Epoch: 10 [51200/60000 (85%)]     Loss: 0.099388
Train Epoch: 10 Average loss: 0.095920
```

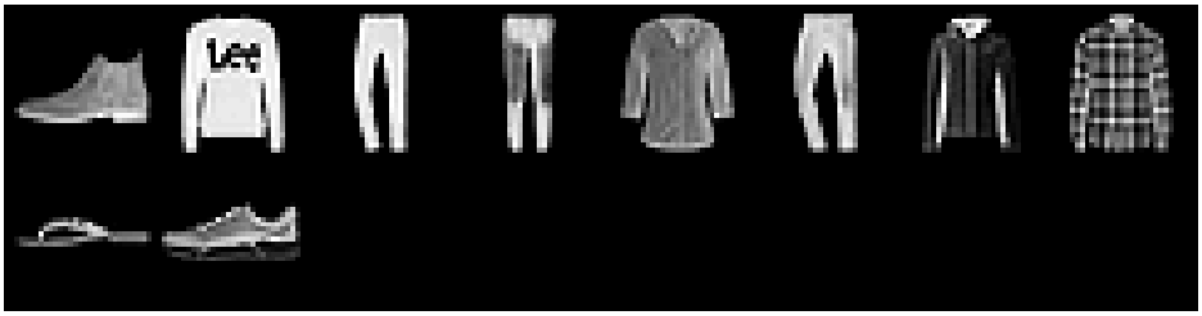
Noisy input



Reconstruction



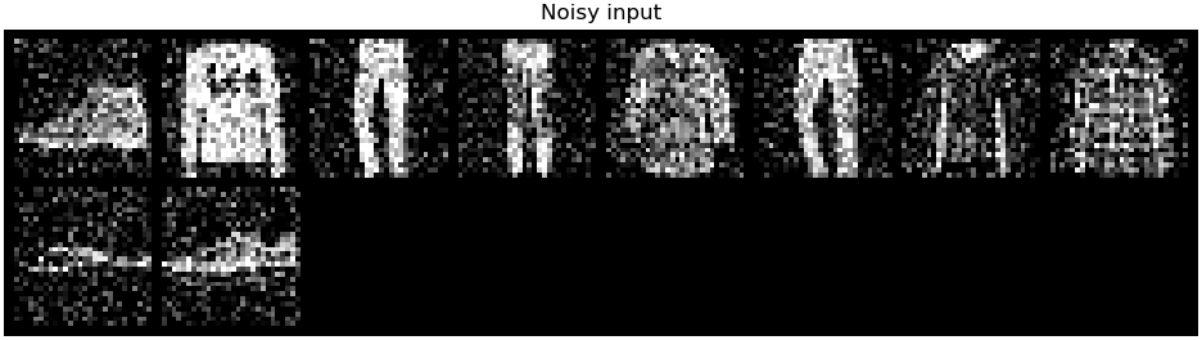
Target



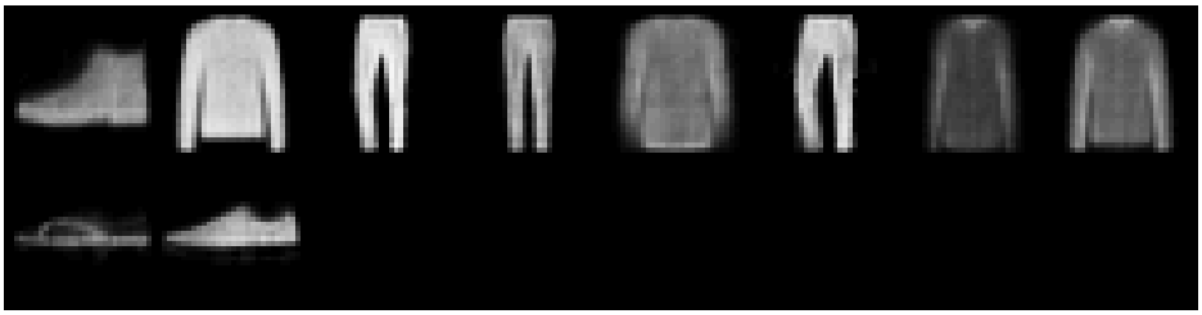
```
Test set: Average loss: 74.9033

Train Epoch: 11 [0/60000 (0%)]  Loss: 0.096699
Train Epoch: 11 [12800/60000 (21%)]     Loss: 0.096911
Train Epoch: 11 [25600/60000 (43%)]     Loss: 0.093033
Train Epoch: 11 [38400/60000 (64%)]     Loss: 0.099041
Train Epoch: 11 [51200/60000 (85%)]     Loss: 0.094803
Train Epoch: 11 Average loss: 0.094964

Test set: Average loss: 74.0908

Train Epoch: 12 [0/60000 (0%)]  Loss: 0.086641
Train Epoch: 12 [12800/60000 (21%)]     Loss: 0.090668
Train Epoch: 12 [25600/60000 (43%)]     Loss: 0.094483
Train Epoch: 12 [38400/60000 (64%)]     Loss: 0.094078
Train Epoch: 12 [51200/60000 (85%)]     Loss: 0.089689
Train Epoch: 12 Average loss: 0.094030

Test set: Average loss: 73.6318

Train Epoch: 13 [0/60000 (0%)]  Loss: 0.089847
Train Epoch: 13 [12800/60000 (21%)]     Loss: 0.098143
Train Epoch: 13 [25600/60000 (43%)]     Loss: 0.093338
Train Epoch: 13 [38400/60000 (64%)]     Loss: 0.091932
Train Epoch: 13 [51200/60000 (85%)]     Loss: 0.085244
Train Epoch: 13 Average loss: 0.093331

Test set: Average loss: 72.8263

Train Epoch: 14 [0/60000 (0%)]  Loss: 0.095019
Train Epoch: 14 [12800/60000 (21%)]     Loss: 0.095499
Train Epoch: 14 [25600/60000 (43%)]     Loss: 0.097458
Train Epoch: 14 [38400/60000 (64%)]     Loss: 0.095640
Train Epoch: 14 [51200/60000 (85%)]     Loss: 0.091158
Train Epoch: 14 Average loss: 0.092623

Test set: Average loss: 72.9550

Train Epoch: 15 [0/60000 (0%)]  Loss: 0.090027
Train Epoch: 15 [12800/60000 (21%)]     Loss: 0.099686
Train Epoch: 15 [25600/60000 (43%)]     Loss: 0.084016
Train Epoch: 15 [38400/60000 (64%)]     Loss: 0.090423
Train Epoch: 15 [51200/60000 (85%)]     Loss: 0.085349
Train Epoch: 15 Average loss: 0.092045

Test set: Average loss: 72.3039

Train Epoch: 16 [0/60000 (0%)]  Loss: 0.083426
Train Epoch: 16 [12800/60000 (21%)]     Loss: 0.093359
Train Epoch: 16 [25600/60000 (43%)]     Loss: 0.090381
Train Epoch: 16 [38400/60000 (64%)]     Loss: 0.081524
Train Epoch: 16 [51200/60000 (85%)]     Loss: 0.092869
Train Epoch: 16 Average loss: 0.091487

Test set: Average loss: 71.6706
```

```
Train Epoch: 17 [0/60000 (0%)]  Loss: 0.088078
Train Epoch: 17 [12800/60000 (21%)]     Loss: 0.089456
Train Epoch: 17 [25600/60000 (43%)]     Loss: 0.094470
Train Epoch: 17 [38400/60000 (64%)]     Loss: 0.092149
Train Epoch: 17 [51200/60000 (85%)]     Loss: 0.101852
Train Epoch: 17 Average loss: 0.091011

Test set: Average loss: 71.2796

Train Epoch: 18 [0/60000 (0%)]  Loss: 0.082342
Train Epoch: 18 [12800/60000 (21%)]     Loss: 0.090200
Train Epoch: 18 [25600/60000 (43%)]     Loss: 0.090200
Train Epoch: 18 [38400/60000 (64%)]     Loss: 0.096682
Train Epoch: 18 [51200/60000 (85%)]     Loss: 0.086296
Train Epoch: 18 Average loss: 0.090669

Test set: Average loss: 71.3886

Train Epoch: 19 [0/60000 (0%)]  Loss: 0.092431
Train Epoch: 19 [12800/60000 (21%)]     Loss: 0.089068
Train Epoch: 19 [25600/60000 (43%)]     Loss: 0.095252
Train Epoch: 19 [38400/60000 (64%)]     Loss: 0.094034
Train Epoch: 19 [51200/60000 (85%)]     Loss: 0.090931
Train Epoch: 19 Average loss: 0.090328

Test set: Average loss: 70.8632

Train Epoch: 20 [0/60000 (0%)]  Loss: 0.089298
Train Epoch: 20 [12800/60000 (21%)]     Loss: 0.090966
Train Epoch: 20 [25600/60000 (43%)]     Loss: 0.095566
Train Epoch: 20 [38400/60000 (64%)]     Loss: 0.096636
Train Epoch: 20 [51200/60000 (85%)]     Loss: 0.092669
Train Epoch: 20 Average loss: 0.089784
```

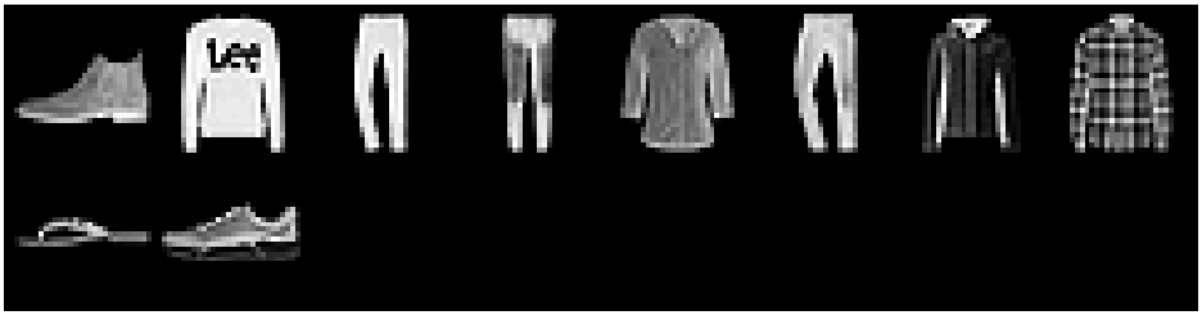
Noisy input



Reconstruction



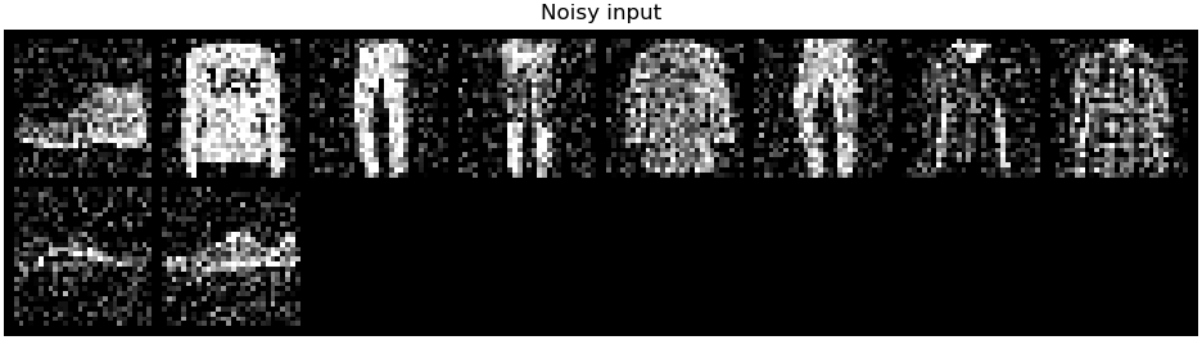
Target



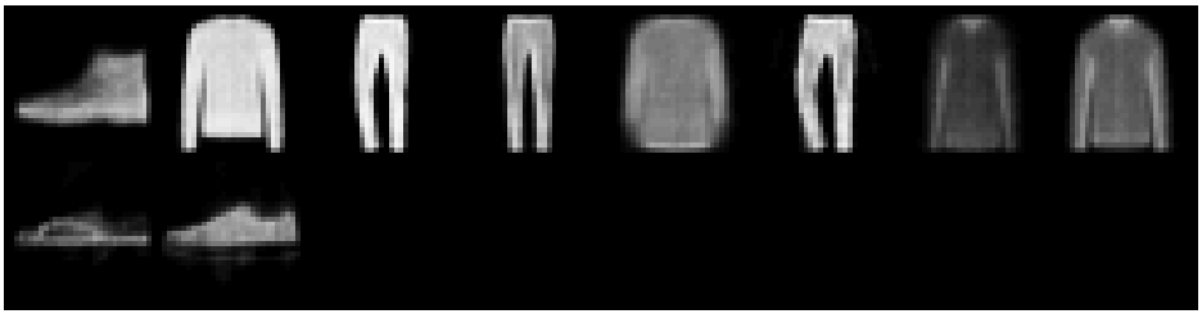
```
Test set: Average loss: 70.4675

Train Epoch: 21 [0/60000 (0%)]  Loss: 0.087446
Train Epoch: 21 [12800/60000 (21%)]     Loss: 0.089531
Train Epoch: 21 [25600/60000 (43%)]     Loss: 0.090167
Train Epoch: 21 [38400/60000 (64%)]     Loss: 0.090054
Train Epoch: 21 [51200/60000 (85%)]     Loss: 0.091028
Train Epoch: 21 Average loss: 0.088312

Test set: Average loss: 69.0263

Train Epoch: 22 [0/60000 (0%)]  Loss: 0.100156
Train Epoch: 22 [12800/60000 (21%)]     Loss: 0.077383
Train Epoch: 22 [25600/60000 (43%)]     Loss: 0.097044
Train Epoch: 22 [38400/60000 (64%)]     Loss: 0.080079
Train Epoch: 22 [51200/60000 (85%)]     Loss: 0.089800
Train Epoch: 22 Average loss: 0.085959

Test set: Average loss: 66.7340

Train Epoch: 23 [0/60000 (0%)]  Loss: 0.084346
Train Epoch: 23 [12800/60000 (21%)]     Loss: 0.084965
Train Epoch: 23 [25600/60000 (43%)]     Loss: 0.084169
Train Epoch: 23 [38400/60000 (64%)]     Loss: 0.082851
Train Epoch: 23 [51200/60000 (85%)]     Loss: 0.090202
Train Epoch: 23 Average loss: 0.084191

Test set: Average loss: 65.8844

Train Epoch: 24 [0/60000 (0%)]  Loss: 0.084356
Train Epoch: 24 [12800/60000 (21%)]     Loss: 0.085648
Train Epoch: 24 [25600/60000 (43%)]     Loss: 0.081195
Train Epoch: 24 [38400/60000 (64%)]     Loss: 0.082935
Train Epoch: 24 [51200/60000 (85%)]     Loss: 0.081714
Train Epoch: 24 Average loss: 0.083302

Test set: Average loss: 65.7459

Train Epoch: 25 [0/60000 (0%)]  Loss: 0.084331
Train Epoch: 25 [12800/60000 (21%)]     Loss: 0.085023
Train Epoch: 25 [25600/60000 (43%)]     Loss: 0.090885
Train Epoch: 25 [38400/60000 (64%)]     Loss: 0.088942
Train Epoch: 25 [51200/60000 (85%)]     Loss: 0.081838
Train Epoch: 25 Average loss: 0.082710

Test set: Average loss: 64.8236

Train Epoch: 26 [0/60000 (0%)]  Loss: 0.082064
Train Epoch: 26 [12800/60000 (21%)]     Loss: 0.078111
Train Epoch: 26 [25600/60000 (43%)]     Loss: 0.079355
Train Epoch: 26 [38400/60000 (64%)]     Loss: 0.077871
Train Epoch: 26 [51200/60000 (85%)]     Loss: 0.078409
Train Epoch: 26 Average loss: 0.082053

Test set: Average loss: 65.0655
```

```
Train Epoch: 27 [0/60000 (0%)]  Loss: 0.086554
Train Epoch: 27 [12800/60000 (21%)]     Loss: 0.085098
Train Epoch: 27 [25600/60000 (43%)]     Loss: 0.083476
Train Epoch: 27 [38400/60000 (64%)]     Loss: 0.082132
Train Epoch: 27 [51200/60000 (85%)]     Loss: 0.081481
Train Epoch: 27 Average loss: 0.081674

Test set: Average loss: 63.9903

Train Epoch: 28 [0/60000 (0%)]  Loss: 0.076441
Train Epoch: 28 [12800/60000 (21%)]     Loss: 0.083808
Train Epoch: 28 [25600/60000 (43%)]     Loss: 0.074990
Train Epoch: 28 [38400/60000 (64%)]     Loss: 0.083430
Train Epoch: 28 [51200/60000 (85%)]     Loss: 0.077495
Train Epoch: 28 Average loss: 0.081268

Test set: Average loss: 63.7653

Train Epoch: 29 [0/60000 (0%)]  Loss: 0.076328
Train Epoch: 29 [12800/60000 (21%)]     Loss: 0.081614
Train Epoch: 29 [25600/60000 (43%)]     Loss: 0.086424
Train Epoch: 29 [38400/60000 (64%)]     Loss: 0.080828
Train Epoch: 29 [51200/60000 (85%)]     Loss: 0.078013
Train Epoch: 29 Average loss: 0.080892

Test set: Average loss: 63.5891

Train Epoch: 30 [0/60000 (0%)]  Loss: 0.073524
Train Epoch: 30 [12800/60000 (21%)]     Loss: 0.083917
Train Epoch: 30 [25600/60000 (43%)]     Loss: 0.083209
Train Epoch: 30 [38400/60000 (64%)]     Loss: 0.085644
Train Epoch: 30 [51200/60000 (85%)]     Loss: 0.088397
Train Epoch: 30 Average loss: 0.080731
```

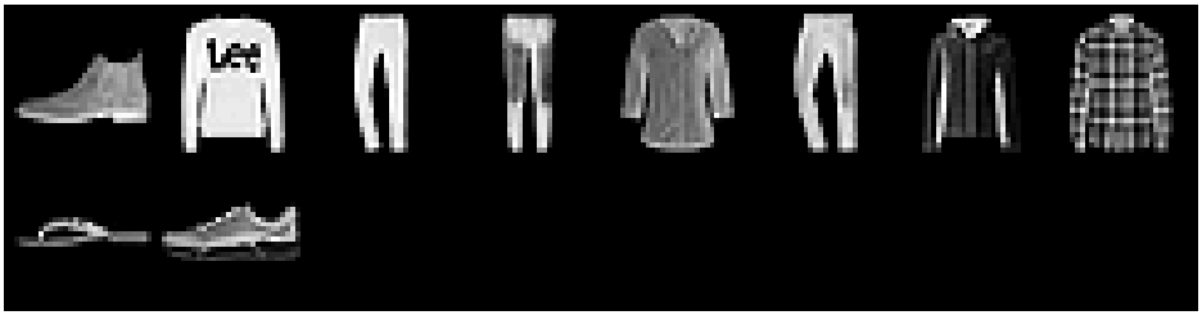
Noisy input



Reconstruction



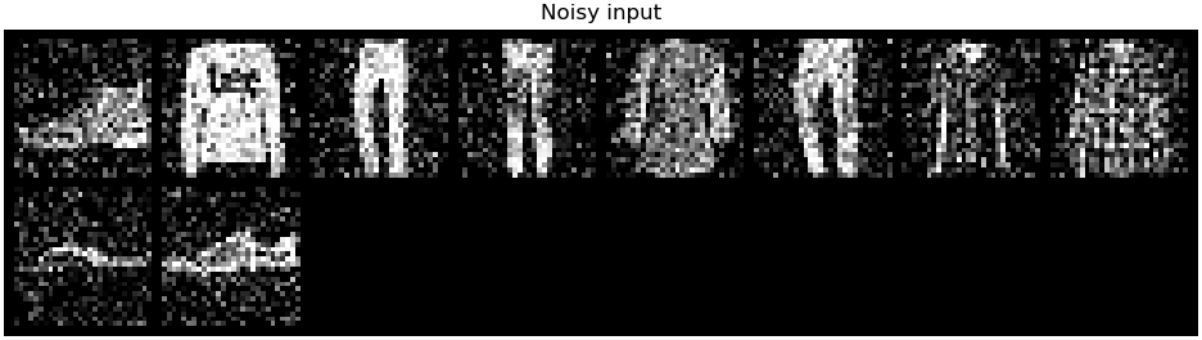
Target



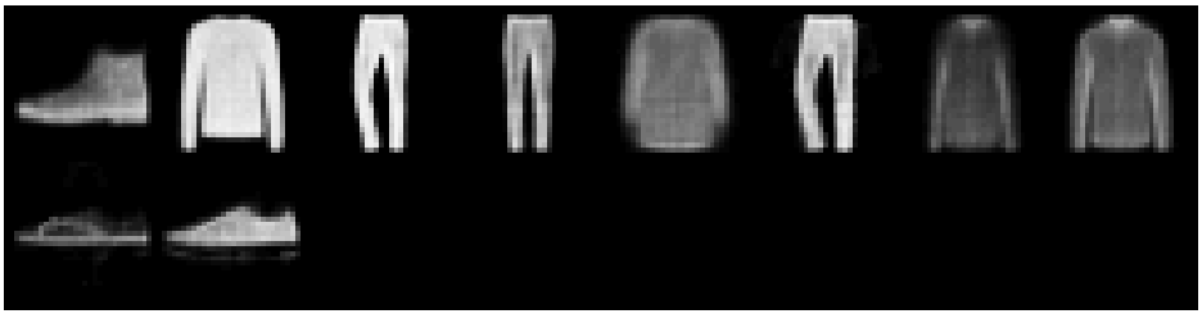
```
Test set: Average loss: 63.3737

Train Epoch: 31 [0/60000 (0%)]  Loss: 0.081379
Train Epoch: 31 [12800/60000 (21%)]     Loss: 0.083595
Train Epoch: 31 [25600/60000 (43%)]     Loss: 0.085769
Train Epoch: 31 [38400/60000 (64%)]     Loss: 0.084153
Train Epoch: 31 [51200/60000 (85%)]     Loss: 0.073114
Train Epoch: 31 Average loss: 0.080409

Test set: Average loss: 63.8410

Train Epoch: 32 [0/60000 (0%)]  Loss: 0.080911
Train Epoch: 32 [12800/60000 (21%)]     Loss: 0.082083
Train Epoch: 32 [25600/60000 (43%)]     Loss: 0.077991
Train Epoch: 32 [38400/60000 (64%)]     Loss: 0.074916
Train Epoch: 32 [51200/60000 (85%)]     Loss: 0.082196
Train Epoch: 32 Average loss: 0.080066

Test set: Average loss: 63.0774

Train Epoch: 33 [0/60000 (0%)]  Loss: 0.080524
Train Epoch: 33 [12800/60000 (21%)]     Loss: 0.086332
Train Epoch: 33 [25600/60000 (43%)]     Loss: 0.075134
Train Epoch: 33 [38400/60000 (64%)]     Loss: 0.087264
Train Epoch: 33 [51200/60000 (85%)]     Loss: 0.077809
Train Epoch: 33 Average loss: 0.079968

Test set: Average loss: 63.7031

Train Epoch: 34 [0/60000 (0%)]  Loss: 0.078321
Train Epoch: 34 [12800/60000 (21%)]     Loss: 0.082242
Train Epoch: 34 [25600/60000 (43%)]     Loss: 0.084278
Train Epoch: 34 [38400/60000 (64%)]     Loss: 0.077332
Train Epoch: 34 [51200/60000 (85%)]     Loss: 0.086915
Train Epoch: 34 Average loss: 0.079649

Test set: Average loss: 63.2284

Train Epoch: 35 [0/60000 (0%)]  Loss: 0.073997
Train Epoch: 35 [12800/60000 (21%)]     Loss: 0.086089
Train Epoch: 35 [25600/60000 (43%)]     Loss: 0.085027
Train Epoch: 35 [38400/60000 (64%)]     Loss: 0.084023
Train Epoch: 35 [51200/60000 (85%)]     Loss: 0.087898
Train Epoch: 35 Average loss: 0.079603

Test set: Average loss: 62.5274

Train Epoch: 36 [0/60000 (0%)]  Loss: 0.080450
Train Epoch: 36 [12800/60000 (21%)]     Loss: 0.073909
Train Epoch: 36 [25600/60000 (43%)]     Loss: 0.082088
Train Epoch: 36 [38400/60000 (64%)]     Loss: 0.077179
Train Epoch: 36 [51200/60000 (85%)]     Loss: 0.082021
Train Epoch: 36 Average loss: 0.079463

Test set: Average loss: 62.2516
```

```
Train Epoch: 37 [0/60000 (0%)]  Loss: 0.080842
Train Epoch: 37 [12800/60000 (21%)]     Loss: 0.081962
Train Epoch: 37 [25600/60000 (43%)]     Loss: 0.076823
Train Epoch: 37 [38400/60000 (64%)]     Loss: 0.075578
Train Epoch: 37 [51200/60000 (85%)]     Loss: 0.075933
Train Epoch: 37 Average loss: 0.079201

Test set: Average loss: 62.5886

Train Epoch: 38 [0/60000 (0%)]  Loss: 0.079141
Train Epoch: 38 [12800/60000 (21%)]     Loss: 0.084481
Train Epoch: 38 [25600/60000 (43%)]     Loss: 0.082386
Train Epoch: 38 [38400/60000 (64%)]     Loss: 0.077323
Train Epoch: 38 [51200/60000 (85%)]     Loss: 0.080147
Train Epoch: 38 Average loss: 0.079048

Test set: Average loss: 62.3912

Train Epoch: 39 [0/60000 (0%)]  Loss: 0.077428
Train Epoch: 39 [12800/60000 (21%)]     Loss: 0.076595
Train Epoch: 39 [25600/60000 (43%)]     Loss: 0.081406
Train Epoch: 39 [38400/60000 (64%)]     Loss: 0.076929
Train Epoch: 39 [51200/60000 (85%)]     Loss: 0.077963
Train Epoch: 39 Average loss: 0.078886

Test set: Average loss: 62.4720

Train Epoch: 40 [0/60000 (0%)]  Loss: 0.078116
Train Epoch: 40 [12800/60000 (21%)]     Loss: 0.080855
Train Epoch: 40 [25600/60000 (43%)]     Loss: 0.080325
Train Epoch: 40 [38400/60000 (64%)]     Loss: 0.087506
Train Epoch: 40 [51200/60000 (85%)]     Loss: 0.079113
Train Epoch: 40 Average loss: 0.078565
```

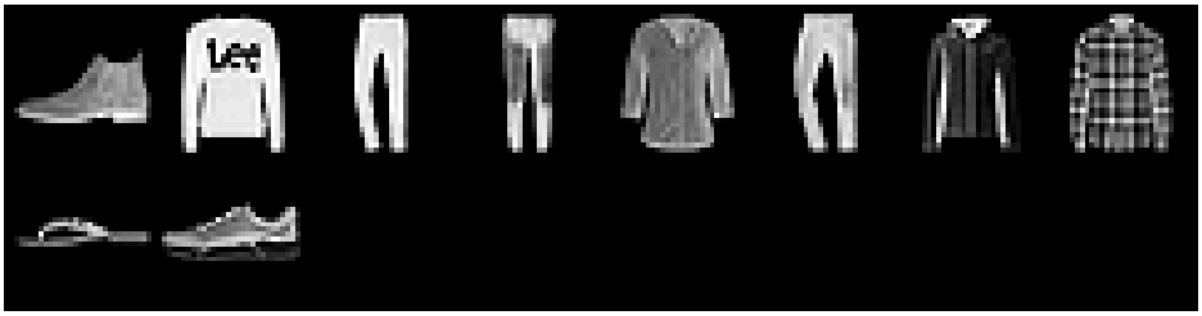
Noisy input



Reconstruction



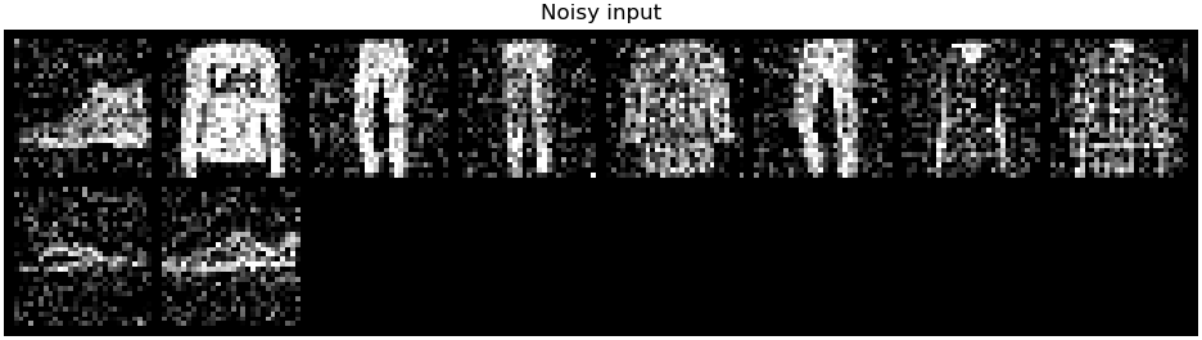
Target



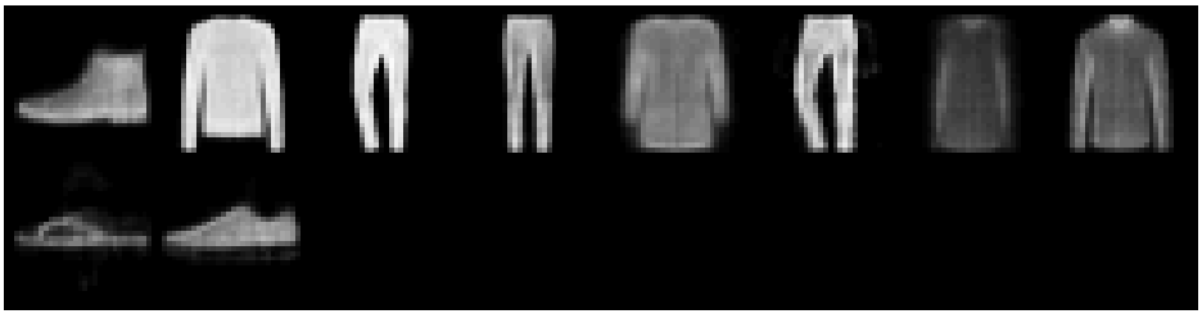
```
Test set: Average loss: 61.9904

Train Epoch: 41 [0/60000 (0%)]  Loss: 0.072168
Train Epoch: 41 [12800/60000 (21%)]     Loss: 0.080962
Train Epoch: 41 [25600/60000 (43%)]     Loss: 0.069428
Train Epoch: 41 [38400/60000 (64%)]     Loss: 0.074757
Train Epoch: 41 [51200/60000 (85%)]     Loss: 0.078903
Train Epoch: 41 Average loss: 0.078450

Test set: Average loss: 61.9229

Train Epoch: 42 [0/60000 (0%)]  Loss: 0.071410
Train Epoch: 42 [12800/60000 (21%)]     Loss: 0.080874
Train Epoch: 42 [25600/60000 (43%)]     Loss: 0.078306
Train Epoch: 42 [38400/60000 (64%)]     Loss: 0.091752
Train Epoch: 42 [51200/60000 (85%)]     Loss: 0.086349
Train Epoch: 42 Average loss: 0.078458

Test set: Average loss: 61.8874

Train Epoch: 43 [0/60000 (0%)]  Loss: 0.078657
Train Epoch: 43 [12800/60000 (21%)]     Loss: 0.077293
Train Epoch: 43 [25600/60000 (43%)]     Loss: 0.081325
Train Epoch: 43 [38400/60000 (64%)]     Loss: 0.074431
Train Epoch: 43 [51200/60000 (85%)]     Loss: 0.072393
Train Epoch: 43 Average loss: 0.078384

Test set: Average loss: 62.0158

Train Epoch: 44 [0/60000 (0%)]  Loss: 0.078191
Train Epoch: 44 [12800/60000 (21%)]     Loss: 0.071707
Train Epoch: 44 [25600/60000 (43%)]     Loss: 0.072452
Train Epoch: 44 [38400/60000 (64%)]     Loss: 0.081428
Train Epoch: 44 [51200/60000 (85%)]     Loss: 0.079286
Train Epoch: 44 Average loss: 0.078361

Test set: Average loss: 61.3703

Train Epoch: 45 [0/60000 (0%)]  Loss: 0.081089
Train Epoch: 45 [12800/60000 (21%)]     Loss: 0.078748
Train Epoch: 45 [25600/60000 (43%)]     Loss: 0.079318
Train Epoch: 45 [38400/60000 (64%)]     Loss: 0.079880
Train Epoch: 45 [51200/60000 (85%)]     Loss: 0.083386
Train Epoch: 45 Average loss: 0.078165

Test set: Average loss: 61.8435

Train Epoch: 46 [0/60000 (0%)]  Loss: 0.072456
Train Epoch: 46 [12800/60000 (21%)]     Loss: 0.083571
Train Epoch: 46 [25600/60000 (43%)]     Loss: 0.079170
Train Epoch: 46 [38400/60000 (64%)]     Loss: 0.077668
Train Epoch: 46 [51200/60000 (85%)]     Loss: 0.076315
Train Epoch: 46 Average loss: 0.077769

Test set: Average loss: 61.4234
```

```
Train Epoch: 47 [0/60000 (0%)]  Loss: 0.074015
Train Epoch: 47 [12800/60000 (21%)]     Loss: 0.082917
Train Epoch: 47 [25600/60000 (43%)]     Loss: 0.073723
Train Epoch: 47 [38400/60000 (64%)]     Loss: 0.073500
Train Epoch: 47 [51200/60000 (85%)]     Loss: 0.079199
Train Epoch: 47 Average loss: 0.077450

Test set: Average loss: 61.1008

Train Epoch: 48 [0/60000 (0%)]  Loss: 0.074273
Train Epoch: 48 [12800/60000 (21%)]     Loss: 0.078982
Train Epoch: 48 [25600/60000 (43%)]     Loss: 0.071284
Train Epoch: 48 [38400/60000 (64%)]     Loss: 0.073069
Train Epoch: 48 [51200/60000 (85%)]     Loss: 0.074568
Train Epoch: 48 Average loss: 0.076809

Test set: Average loss: 60.6138

Train Epoch: 49 [0/60000 (0%)]  Loss: 0.077868
Train Epoch: 49 [12800/60000 (21%)]     Loss: 0.075064
Train Epoch: 49 [25600/60000 (43%)]     Loss: 0.076263
Train Epoch: 49 [38400/60000 (64%)]     Loss: 0.072458
Train Epoch: 49 [51200/60000 (85%)]     Loss: 0.071589
Train Epoch: 49 Average loss: 0.076064

Test set: Average loss: 59.3761

Train Epoch: 50 [0/60000 (0%)]  Loss: 0.074600
Train Epoch: 50 [12800/60000 (21%)]     Loss: 0.075346
Train Epoch: 50 [25600/60000 (43%)]     Loss: 0.068877
Train Epoch: 50 [38400/60000 (64%)]     Loss: 0.078046
Train Epoch: 50 [51200/60000 (85%)]     Loss: 0.069864
Train Epoch: 50 Average loss: 0.075222
```

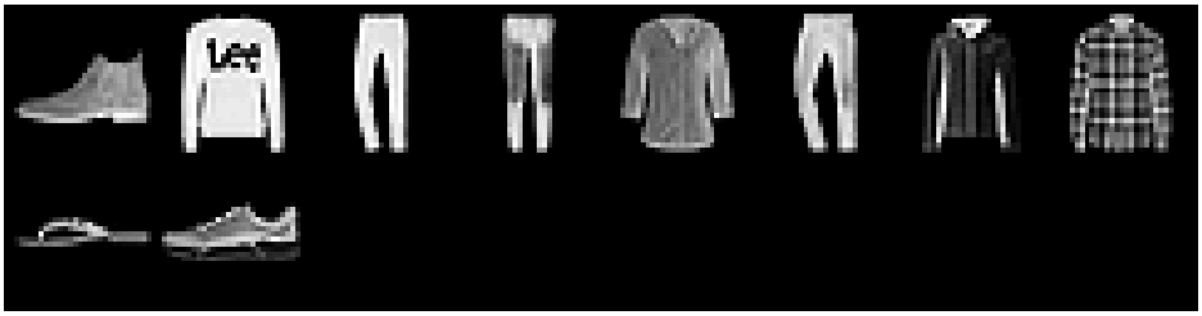
Noisy input



Reconstruction



Target



```
Test set: Average loss: 59.0620

The best (lowest) achieved loss: 59.062
```

Task 2 Written Answer

TODO 2.1: Briefly describe how the denoising autoencoder performs compared with the plain autoencoder. Mention the observed losses and the visual reconstruction quality.

Your answer: The denoising outoencoder performs marginally better than the plain autoencoder (Average loss 59.0620 vs. 59.9906). However, while the visual reconstruction qualtiy is very similar for both versions of the autoencoder the denoising worked and the reconstruction removed the noise and resulting in a blured version of the original images.

Task 3: Convolutional Autoencoder

Implement the `Conv_Autoencoder` class below.

Encoder architecture:

- `Conv2d(1, 4, kernel_size=5)` + `ReLU`
- `Conv2d(4, 8, kernel_size=5)` + `ReLU`
- `Flatten`
- `Linear(3200, 10)`
- `Softmax(dim=1)`

Decoder architecture:

- `Linear(10, 400)` + `ReLU`
- `Linear(400, 4000)` + `ReLU`
- `Unflatten` to `(10, 20, 20)`
- `ConvTranspose2d(10, 10, kernel_size=5)` + `ReLU`
- `ConvTranspose2d(10, 1, kernel_size=5)` + `Tanh`

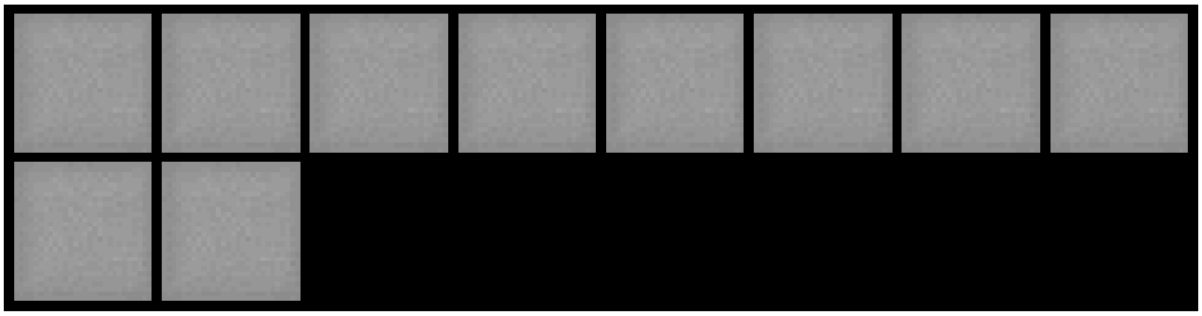
In [23]:
```
class Conv_Autoencoder(nn.Module):
    def __init__(self):
        super(Conv_Autoencoder, self).__init__()

        encoder_layers = [
            nn.Conv2d(1, 4, kernel_size=5),
            nn.ReLU(),
            nn.Conv2d(4, 8, kernel_size=5),
            nn.ReLU(),
            nn.Flatten(),
            nn.Linear(3200, 10),
            nn.Softmax(dim=1)
        ]
        self.encoder = nn.Sequential(*encoder_layers)

        decoder_layers = [
            nn.Linear(10, 400),
            nn.ReLU(),
            nn.Linear(400, 4000),
            nn.ReLU(),
            nn.Unflatten(1, (10, 20, 20)),
            nn.ConvTranspose2d(10, 10, kernel_size=5),
            nn.ReLU(),
            nn.ConvTranspose2d(10, 1, kernel_size=5),
            nn.Tanh(),
        ]
        self.decoder = nn.Sequential(*decoder_layers)

    def forward(self, x):
        x = x.view(-1, 1, 28, 28)
        latend_code = self.encoder(x)
        reconstruction = self.decoder(latend_code)
        return reconstruction.view(x.size(0), -1)
```

In [24]:
```
# Run this after completing Task 3. The output shape must match the flattene
conv_shape_test = Conv_Autoencoder()
with torch.no_grad():
    dummy = torch.randn(4, 28 * 28)
    assert conv_shape_test(dummy).shape == dummy.shape
print('Conv_Autoencoder shape check passed.')
```

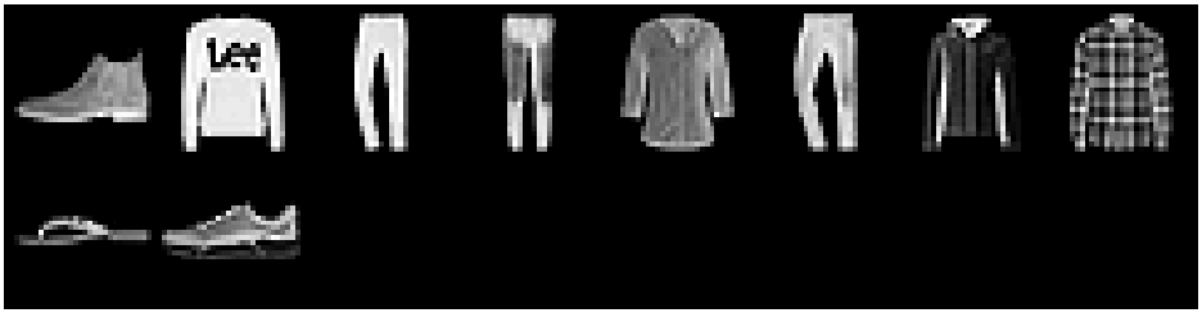
```
Conv_Autoencoder shape check passed.
```

In [25]:
```
# Task 3 training: convolutional autoencoder.
args.is_noisy = False
best_conv_loss = run_experiment(
    Conv_Autoencoder(),
    save_path='best_conv_autoencoder.pth',
    noisy=args.is_noisy,
)
```

Reconstruction



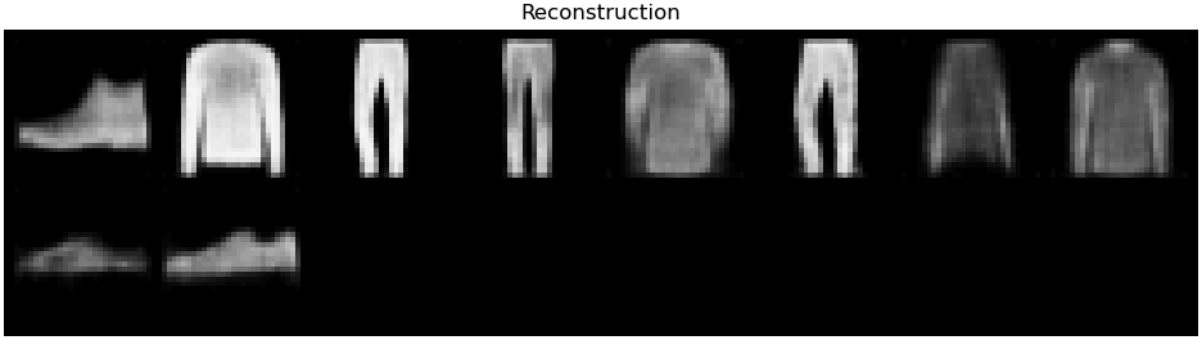
Target



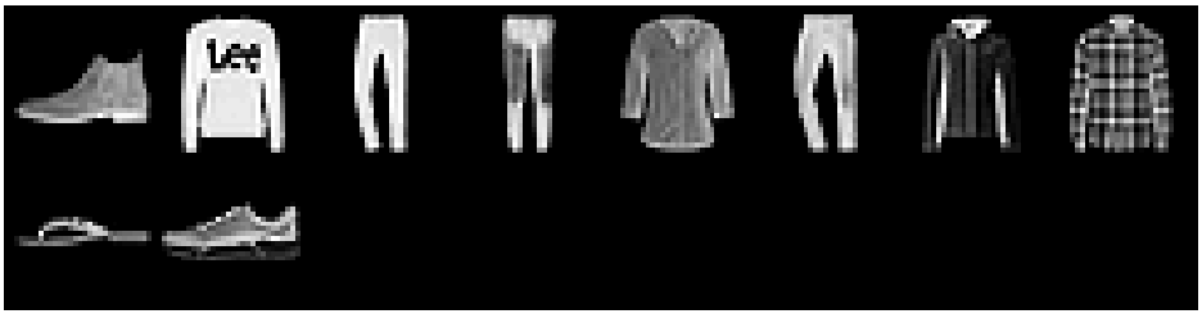
```
Test set: Average loss: 682.9933

Train Epoch: 1 [0/60000 (0%)]   Loss: 0.864436
Train Epoch: 1 [12800/60000 (21%)]      Loss: 0.190494
Train Epoch: 1 [25600/60000 (43%)]      Loss: 0.114902
Train Epoch: 1 [38400/60000 (64%)]      Loss: 0.110240
Train Epoch: 1 [51200/60000 (85%)]      Loss: 0.097171
Train Epoch: 1 Average loss: 0.170379

Test set: Average loss: 72.5296

Train Epoch: 2 [0/60000 (0%)]   Loss: 0.091211
Train Epoch: 2 [12800/60000 (21%)]      Loss: 0.088518
Train Epoch: 2 [25600/60000 (43%)]      Loss: 0.082501
Train Epoch: 2 [38400/60000 (64%)]      Loss: 0.080413
Train Epoch: 2 [51200/60000 (85%)]      Loss: 0.084636
Train Epoch: 2 Average loss: 0.086436

Test set: Average loss: 63.4587

Train Epoch: 3 [0/60000 (0%)]   Loss: 0.083498
Train Epoch: 3 [12800/60000 (21%)]      Loss: 0.088613
Train Epoch: 3 [25600/60000 (43%)]      Loss: 0.081798
Train Epoch: 3 [38400/60000 (64%)]      Loss: 0.078628
Train Epoch: 3 [51200/60000 (85%)]      Loss: 0.074595
Train Epoch: 3 Average loss: 0.078625

Test set: Average loss: 59.5971

Train Epoch: 4 [0/60000 (0%)]   Loss: 0.075562
Train Epoch: 4 [12800/60000 (21%)]      Loss: 0.076062
Train Epoch: 4 [25600/60000 (43%)]      Loss: 0.074075
Train Epoch: 4 [38400/60000 (64%)]      Loss: 0.076842
Train Epoch: 4 [51200/60000 (85%)]      Loss: 0.069245
Train Epoch: 4 Average loss: 0.074004

Test set: Average loss: 56.6802

Train Epoch: 5 [0/60000 (0%)]   Loss: 0.067263
Train Epoch: 5 [12800/60000 (21%)]      Loss: 0.076240
Train Epoch: 5 [25600/60000 (43%)]      Loss: 0.070760
Train Epoch: 5 [38400/60000 (64%)]      Loss: 0.070706
Train Epoch: 5 [51200/60000 (85%)]      Loss: 0.076220
Train Epoch: 5 Average loss: 0.071246

Test set: Average loss: 55.4506

Train Epoch: 6 [0/60000 (0%)]   Loss: 0.071517
Train Epoch: 6 [12800/60000 (21%)]      Loss: 0.062370
Train Epoch: 6 [25600/60000 (43%)]      Loss: 0.072744
Train Epoch: 6 [38400/60000 (64%)]      Loss: 0.071782
Train Epoch: 6 [51200/60000 (85%)]      Loss: 0.069200
Train Epoch: 6 Average loss: 0.069017

Test set: Average loss: 54.0616
```

```
Train Epoch: 7 [0/60000 (0%)]   Loss: 0.068857
Train Epoch: 7 [12800/60000 (21%)]      Loss: 0.073956
Train Epoch: 7 [25600/60000 (43%)]      Loss: 0.063460
Train Epoch: 7 [38400/60000 (64%)]      Loss: 0.071000
Train Epoch: 7 [51200/60000 (85%)]      Loss: 0.061863
Train Epoch: 7 Average loss: 0.067182

Test set: Average loss: 52.3289

Train Epoch: 8 [0/60000 (0%)]   Loss: 0.065682
Train Epoch: 8 [12800/60000 (21%)]      Loss: 0.069778
Train Epoch: 8 [25600/60000 (43%)]      Loss: 0.065764
Train Epoch: 8 [38400/60000 (64%)]      Loss: 0.066055
Train Epoch: 8 [51200/60000 (85%)]      Loss: 0.062495
Train Epoch: 8 Average loss: 0.065717

Test set: Average loss: 51.2230

Train Epoch: 9 [0/60000 (0%)]   Loss: 0.067103
Train Epoch: 9 [12800/60000 (21%)]      Loss: 0.069545
Train Epoch: 9 [25600/60000 (43%)]      Loss: 0.063371
Train Epoch: 9 [38400/60000 (64%)]      Loss: 0.067103
Train Epoch: 9 [51200/60000 (85%)]      Loss: 0.061911
Train Epoch: 9 Average loss: 0.064258

Test set: Average loss: 50.2586

Train Epoch: 10 [0/60000 (0%)]  Loss: 0.063044
Train Epoch: 10 [12800/60000 (21%)]     Loss: 0.069544
Train Epoch: 10 [25600/60000 (43%)]     Loss: 0.062454
Train Epoch: 10 [38400/60000 (64%)]     Loss: 0.064797
Train Epoch: 10 [51200/60000 (85%)]     Loss: 0.057712
Train Epoch: 10 Average loss: 0.063039
```

Reconstruction



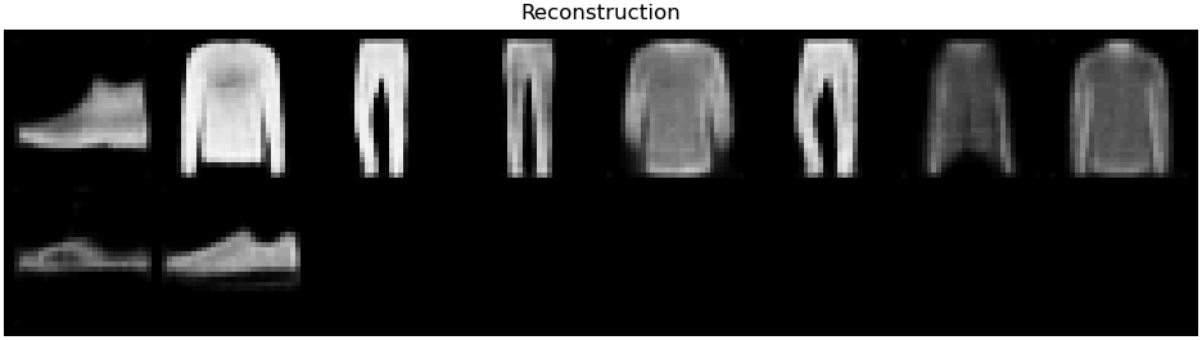
Target



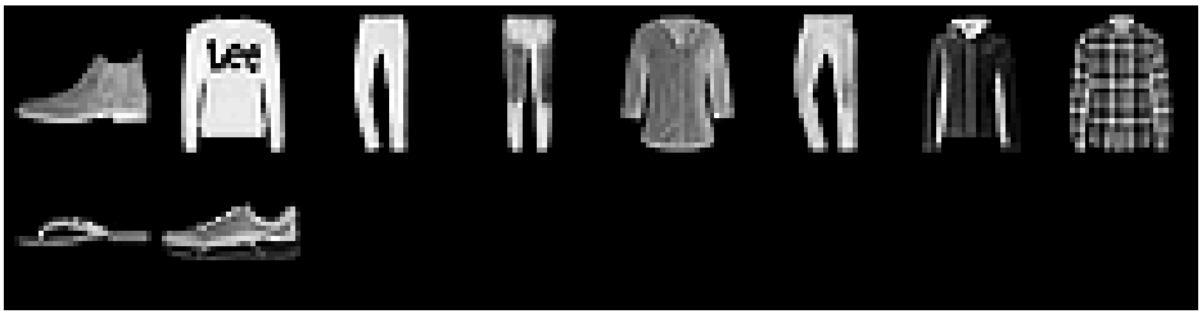
```
Test set: Average loss: 49.1460

Train Epoch: 11 [0/60000 (0%)]  Loss: 0.060998
Train Epoch: 11 [12800/60000 (21%)]     Loss: 0.065151
Train Epoch: 11 [25600/60000 (43%)]     Loss: 0.062631
Train Epoch: 11 [38400/60000 (64%)]     Loss: 0.055995
Train Epoch: 11 [51200/60000 (85%)]     Loss: 0.062957
Train Epoch: 11 Average loss: 0.062009

Test set: Average loss: 49.3139

Train Epoch: 12 [0/60000 (0%)]  Loss: 0.059408
Train Epoch: 12 [12800/60000 (21%)]     Loss: 0.051121
Train Epoch: 12 [25600/60000 (43%)]     Loss: 0.063347
Train Epoch: 12 [38400/60000 (64%)]     Loss: 0.067154
Train Epoch: 12 [51200/60000 (85%)]     Loss: 0.059706
Train Epoch: 12 Average loss: 0.061117

Test set: Average loss: 47.7320

Train Epoch: 13 [0/60000 (0%)]  Loss: 0.058009
Train Epoch: 13 [12800/60000 (21%)]     Loss: 0.060902
Train Epoch: 13 [25600/60000 (43%)]     Loss: 0.066688
Train Epoch: 13 [38400/60000 (64%)]     Loss: 0.057234
Train Epoch: 13 [51200/60000 (85%)]     Loss: 0.063114
Train Epoch: 13 Average loss: 0.060154

Test set: Average loss: 47.2426

Train Epoch: 14 [0/60000 (0%)]  Loss: 0.058325
Train Epoch: 14 [12800/60000 (21%)]     Loss: 0.057834
Train Epoch: 14 [25600/60000 (43%)]     Loss: 0.062365
Train Epoch: 14 [38400/60000 (64%)]     Loss: 0.053523
Train Epoch: 14 [51200/60000 (85%)]     Loss: 0.060154
Train Epoch: 14 Average loss: 0.059442

Test set: Average loss: 46.5374

Train Epoch: 15 [0/60000 (0%)]  Loss: 0.054951
Train Epoch: 15 [12800/60000 (21%)]     Loss: 0.063185
Train Epoch: 15 [25600/60000 (43%)]     Loss: 0.055862
Train Epoch: 15 [38400/60000 (64%)]     Loss: 0.058913
Train Epoch: 15 [51200/60000 (85%)]     Loss: 0.055889
Train Epoch: 15 Average loss: 0.058621

Test set: Average loss: 46.9341

Train Epoch: 16 [0/60000 (0%)]  Loss: 0.061033
Train Epoch: 16 [12800/60000 (21%)]     Loss: 0.055727
Train Epoch: 16 [25600/60000 (43%)]     Loss: 0.053061
Train Epoch: 16 [38400/60000 (64%)]     Loss: 0.057817
Train Epoch: 16 [51200/60000 (85%)]     Loss: 0.059687
Train Epoch: 16 Average loss: 0.058046

Test set: Average loss: 46.3069
```

```
Train Epoch: 17 [0/60000 (0%)]  Loss: 0.053053
Train Epoch: 17 [12800/60000 (21%)]     Loss: 0.056116
Train Epoch: 17 [25600/60000 (43%)]     Loss: 0.058168
Train Epoch: 17 [38400/60000 (64%)]     Loss: 0.052756
Train Epoch: 17 [51200/60000 (85%)]     Loss: 0.058812
Train Epoch: 17 Average loss: 0.057370

Test set: Average loss: 45.4169

Train Epoch: 18 [0/60000 (0%)]  Loss: 0.053995
Train Epoch: 18 [12800/60000 (21%)]     Loss: 0.055709
Train Epoch: 18 [25600/60000 (43%)]     Loss: 0.060217
Train Epoch: 18 [38400/60000 (64%)]     Loss: 0.060340
Train Epoch: 18 [51200/60000 (85%)]     Loss: 0.057145
Train Epoch: 18 Average loss: 0.057031

Test set: Average loss: 45.3752

Train Epoch: 19 [0/60000 (0%)]  Loss: 0.057643
Train Epoch: 19 [12800/60000 (21%)]     Loss: 0.062313
Train Epoch: 19 [25600/60000 (43%)]     Loss: 0.059368
Train Epoch: 19 [38400/60000 (64%)]     Loss: 0.054699
Train Epoch: 19 [51200/60000 (85%)]     Loss: 0.054968
Train Epoch: 19 Average loss: 0.056448

Test set: Average loss: 44.6817

Train Epoch: 20 [0/60000 (0%)]  Loss: 0.056868
Train Epoch: 20 [12800/60000 (21%)]     Loss: 0.054682
Train Epoch: 20 [25600/60000 (43%)]     Loss: 0.052546
Train Epoch: 20 [38400/60000 (64%)]     Loss: 0.051763
Train Epoch: 20 [51200/60000 (85%)]     Loss: 0.051139
Train Epoch: 20 Average loss: 0.055928
```

Reconstruction



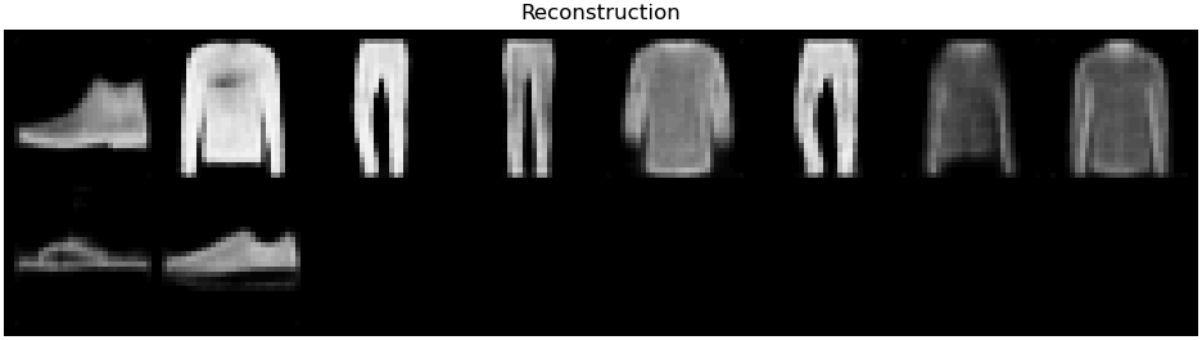
Target



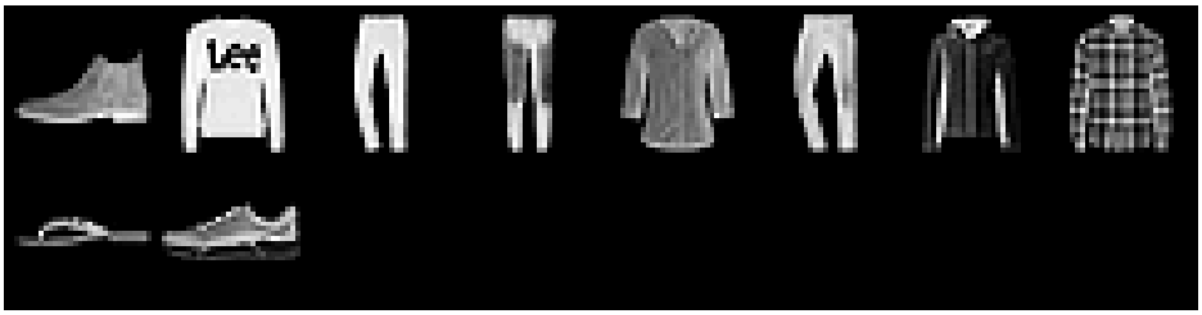
```
Test set: Average loss: 44.0524

Train Epoch: 21 [0/60000 (0%)]  Loss: 0.053268
Train Epoch: 21 [12800/60000 (21%)]     Loss: 0.058552
Train Epoch: 21 [25600/60000 (43%)]     Loss: 0.050551
Train Epoch: 21 [38400/60000 (64%)]     Loss: 0.046225
Train Epoch: 21 [51200/60000 (85%)]     Loss: 0.054658
Train Epoch: 21 Average loss: 0.055480

Test set: Average loss: 44.0462

Train Epoch: 22 [0/60000 (0%)]  Loss: 0.061282
Train Epoch: 22 [12800/60000 (21%)]     Loss: 0.053310
Train Epoch: 22 [25600/60000 (43%)]     Loss: 0.052182
Train Epoch: 22 [38400/60000 (64%)]     Loss: 0.052531
Train Epoch: 22 [51200/60000 (85%)]     Loss: 0.055873
Train Epoch: 22 Average loss: 0.055151

Test set: Average loss: 44.0184

Train Epoch: 23 [0/60000 (0%)]  Loss: 0.050865
Train Epoch: 23 [12800/60000 (21%)]     Loss: 0.051059
Train Epoch: 23 [25600/60000 (43%)]     Loss: 0.056417
Train Epoch: 23 [38400/60000 (64%)]     Loss: 0.055200
Train Epoch: 23 [51200/60000 (85%)]     Loss: 0.053582
Train Epoch: 23 Average loss: 0.054733

Test set: Average loss: 43.4309

Train Epoch: 24 [0/60000 (0%)]  Loss: 0.055665
Train Epoch: 24 [12800/60000 (21%)]     Loss: 0.056777
Train Epoch: 24 [25600/60000 (43%)]     Loss: 0.051802
Train Epoch: 24 [38400/60000 (64%)]     Loss: 0.057546
Train Epoch: 24 [51200/60000 (85%)]     Loss: 0.055753
Train Epoch: 24 Average loss: 0.054480

Test set: Average loss: 43.2429

Train Epoch: 25 [0/60000 (0%)]  Loss: 0.052066
Train Epoch: 25 [12800/60000 (21%)]     Loss: 0.053228
Train Epoch: 25 [25600/60000 (43%)]     Loss: 0.057841
Train Epoch: 25 [38400/60000 (64%)]     Loss: 0.054702
Train Epoch: 25 [51200/60000 (85%)]     Loss: 0.050470
Train Epoch: 25 Average loss: 0.054074

Test set: Average loss: 43.1376

Train Epoch: 26 [0/60000 (0%)]  Loss: 0.051995
Train Epoch: 26 [12800/60000 (21%)]     Loss: 0.055969
Train Epoch: 26 [25600/60000 (43%)]     Loss: 0.045685
Train Epoch: 26 [38400/60000 (64%)]     Loss: 0.051789
Train Epoch: 26 [51200/60000 (85%)]     Loss: 0.054077
Train Epoch: 26 Average loss: 0.053912

Test set: Average loss: 42.8027
```

```
Train Epoch: 27 [0/60000 (0%)]  Loss: 0.055419
Train Epoch: 27 [12800/60000 (21%)]     Loss: 0.055503
Train Epoch: 27 [25600/60000 (43%)]     Loss: 0.049155
Train Epoch: 27 [38400/60000 (64%)]     Loss: 0.047654
Train Epoch: 27 [51200/60000 (85%)]     Loss: 0.060603
Train Epoch: 27 Average loss: 0.053686

Test set: Average loss: 42.5463

Train Epoch: 28 [0/60000 (0%)]  Loss: 0.048582
Train Epoch: 28 [12800/60000 (21%)]     Loss: 0.051804
Train Epoch: 28 [25600/60000 (43%)]     Loss: 0.054888
Train Epoch: 28 [38400/60000 (64%)]     Loss: 0.049147
Train Epoch: 28 [51200/60000 (85%)]     Loss: 0.053419
Train Epoch: 28 Average loss: 0.053391

Test set: Average loss: 42.2310

Train Epoch: 29 [0/60000 (0%)]  Loss: 0.053133
Train Epoch: 29 [12800/60000 (21%)]     Loss: 0.055477
Train Epoch: 29 [25600/60000 (43%)]     Loss: 0.056570
Train Epoch: 29 [38400/60000 (64%)]     Loss: 0.049003
Train Epoch: 29 [51200/60000 (85%)]     Loss: 0.057309
Train Epoch: 29 Average loss: 0.053219

Test set: Average loss: 42.1029

Train Epoch: 30 [0/60000 (0%)]  Loss: 0.056080
Train Epoch: 30 [12800/60000 (21%)]     Loss: 0.051993
Train Epoch: 30 [25600/60000 (43%)]     Loss: 0.053058
Train Epoch: 30 [38400/60000 (64%)]     Loss: 0.054806
Train Epoch: 30 [51200/60000 (85%)]     Loss: 0.050163
Train Epoch: 30 Average loss: 0.052996
```

Reconstruction



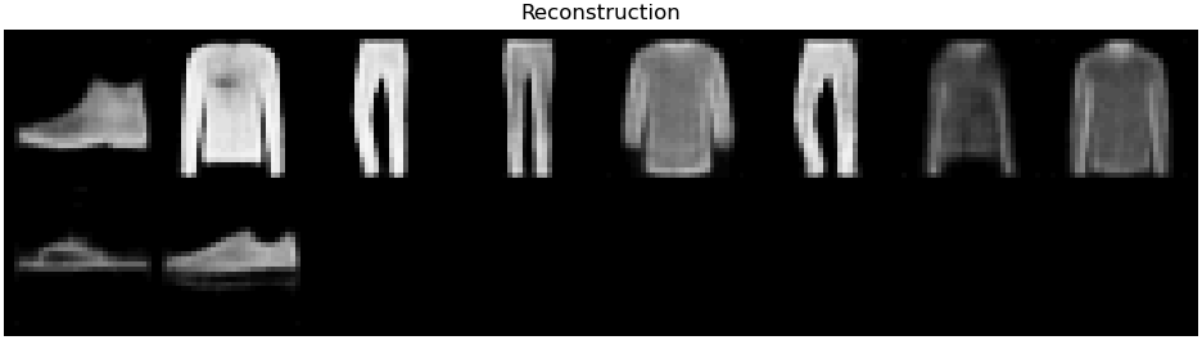
Target



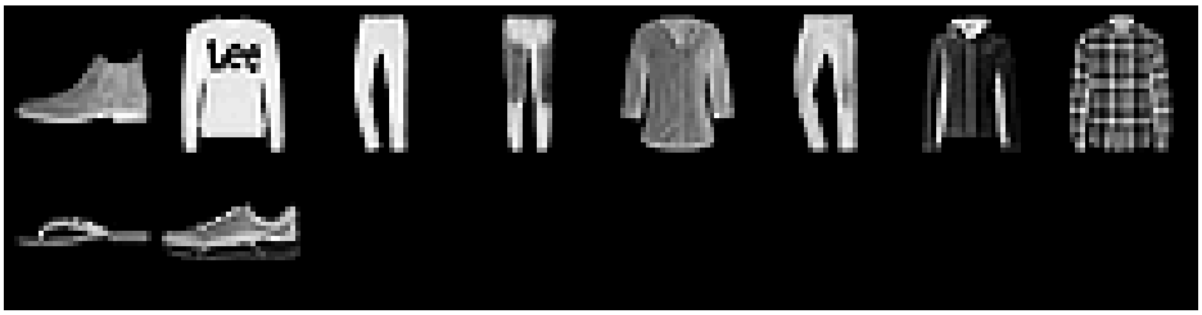
```
Test set: Average loss: 41.9400

Train Epoch: 31 [0/60000 (0%)]  Loss: 0.052655
Train Epoch: 31 [12800/60000 (21%)]     Loss: 0.055789
Train Epoch: 31 [25600/60000 (43%)]     Loss: 0.048808
Train Epoch: 31 [38400/60000 (64%)]     Loss: 0.054584
Train Epoch: 31 [51200/60000 (85%)]     Loss: 0.054113
Train Epoch: 31 Average loss: 0.052796

Test set: Average loss: 41.8682

Train Epoch: 32 [0/60000 (0%)]  Loss: 0.050558
Train Epoch: 32 [12800/60000 (21%)]     Loss: 0.056949
Train Epoch: 32 [25600/60000 (43%)]     Loss: 0.051939
Train Epoch: 32 [38400/60000 (64%)]     Loss: 0.055014
Train Epoch: 32 [51200/60000 (85%)]     Loss: 0.050709
Train Epoch: 32 Average loss: 0.052541

Test set: Average loss: 42.1858

Train Epoch: 33 [0/60000 (0%)]  Loss: 0.054073
Train Epoch: 33 [12800/60000 (21%)]     Loss: 0.049813
Train Epoch: 33 [25600/60000 (43%)]     Loss: 0.047973
Train Epoch: 33 [38400/60000 (64%)]     Loss: 0.048637
Train Epoch: 33 [51200/60000 (85%)]     Loss: 0.059194
Train Epoch: 33 Average loss: 0.052494

Test set: Average loss: 41.6425

Train Epoch: 34 [0/60000 (0%)]  Loss: 0.049850
Train Epoch: 34 [12800/60000 (21%)]     Loss: 0.056666
Train Epoch: 34 [25600/60000 (43%)]     Loss: 0.054895
Train Epoch: 34 [38400/60000 (64%)]     Loss: 0.048119
Train Epoch: 34 [51200/60000 (85%)]     Loss: 0.053982
Train Epoch: 34 Average loss: 0.052282

Test set: Average loss: 41.4546

Train Epoch: 35 [0/60000 (0%)]  Loss: 0.053589
Train Epoch: 35 [12800/60000 (21%)]     Loss: 0.048411
Train Epoch: 35 [25600/60000 (43%)]     Loss: 0.046639
Train Epoch: 35 [38400/60000 (64%)]     Loss: 0.051617
Train Epoch: 35 [51200/60000 (85%)]     Loss: 0.053807
Train Epoch: 35 Average loss: 0.052156

Test set: Average loss: 41.7841

Train Epoch: 36 [0/60000 (0%)]  Loss: 0.050462
Train Epoch: 36 [12800/60000 (21%)]     Loss: 0.048946
Train Epoch: 36 [25600/60000 (43%)]     Loss: 0.055923
Train Epoch: 36 [38400/60000 (64%)]     Loss: 0.052297
Train Epoch: 36 [51200/60000 (85%)]     Loss: 0.051872
Train Epoch: 36 Average loss: 0.052018

Test set: Average loss: 41.7115
```

```
Train Epoch: 37 [0/60000 (0%)]  Loss: 0.049666
Train Epoch: 37 [12800/60000 (21%)]     Loss: 0.054190
Train Epoch: 37 [25600/60000 (43%)]     Loss: 0.055383
Train Epoch: 37 [38400/60000 (64%)]     Loss: 0.051221
Train Epoch: 37 [51200/60000 (85%)]     Loss: 0.046302
Train Epoch: 37 Average loss: 0.051984

Test set: Average loss: 41.2409

Train Epoch: 38 [0/60000 (0%)]  Loss: 0.055837
Train Epoch: 38 [12800/60000 (21%)]     Loss: 0.048421
Train Epoch: 38 [25600/60000 (43%)]     Loss: 0.053098
Train Epoch: 38 [38400/60000 (64%)]     Loss: 0.052902
Train Epoch: 38 [51200/60000 (85%)]     Loss: 0.053001
Train Epoch: 38 Average loss: 0.051728

Test set: Average loss: 41.3529

Train Epoch: 39 [0/60000 (0%)]  Loss: 0.041854
Train Epoch: 39 [12800/60000 (21%)]     Loss: 0.051897
Train Epoch: 39 [25600/60000 (43%)]     Loss: 0.051523
Train Epoch: 39 [38400/60000 (64%)]     Loss: 0.045986
Train Epoch: 39 [51200/60000 (85%)]     Loss: 0.050456
Train Epoch: 39 Average loss: 0.051621

Test set: Average loss: 41.4814

Train Epoch: 40 [0/60000 (0%)]  Loss: 0.055274
Train Epoch: 40 [12800/60000 (21%)]     Loss: 0.045179
Train Epoch: 40 [25600/60000 (43%)]     Loss: 0.047422
Train Epoch: 40 [38400/60000 (64%)]     Loss: 0.049813
Train Epoch: 40 [51200/60000 (85%)]     Loss: 0.053591
Train Epoch: 40 Average loss: 0.051571
```

Reconstruction



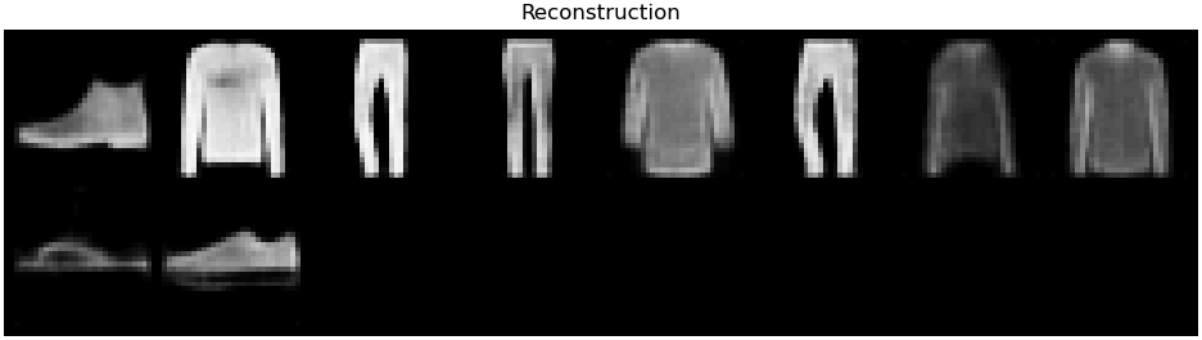
Target



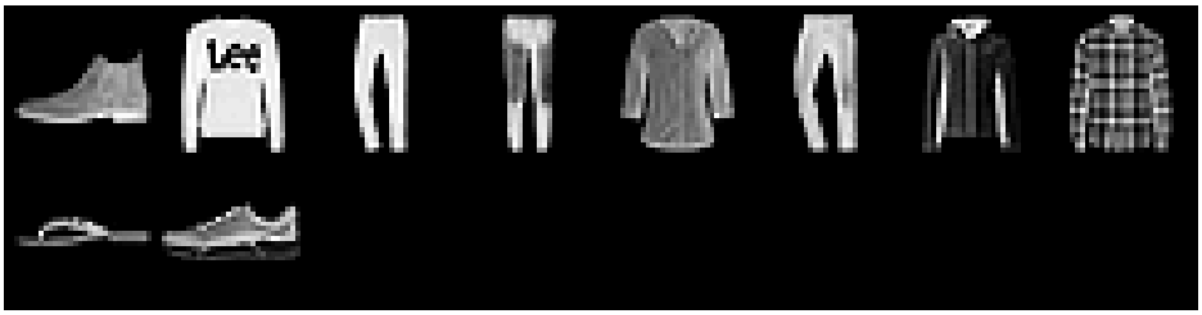
```
Test set: Average loss: 41.1154

Train Epoch: 41 [0/60000 (0%)]  Loss: 0.049031
Train Epoch: 41 [12800/60000 (21%)]     Loss: 0.049071
Train Epoch: 41 [25600/60000 (43%)]     Loss: 0.051498
Train Epoch: 41 [38400/60000 (64%)]     Loss: 0.047194
Train Epoch: 41 [51200/60000 (85%)]     Loss: 0.054586
Train Epoch: 41 Average loss: 0.051467

Test set: Average loss: 40.7409

Train Epoch: 42 [0/60000 (0%)]  Loss: 0.051907
Train Epoch: 42 [12800/60000 (21%)]     Loss: 0.052932
Train Epoch: 42 [25600/60000 (43%)]     Loss: 0.056008
Train Epoch: 42 [38400/60000 (64%)]     Loss: 0.053988
Train Epoch: 42 [51200/60000 (85%)]     Loss: 0.047935
Train Epoch: 42 Average loss: 0.051382

Test set: Average loss: 40.9201

Train Epoch: 43 [0/60000 (0%)]  Loss: 0.052719
Train Epoch: 43 [12800/60000 (21%)]     Loss: 0.054053
Train Epoch: 43 [25600/60000 (43%)]     Loss: 0.049777
Train Epoch: 43 [38400/60000 (64%)]     Loss: 0.063259
Train Epoch: 43 [51200/60000 (85%)]     Loss: 0.049051
Train Epoch: 43 Average loss: 0.051239

Test set: Average loss: 41.0492

Train Epoch: 44 [0/60000 (0%)]  Loss: 0.059103
Train Epoch: 44 [12800/60000 (21%)]     Loss: 0.049877
Train Epoch: 44 [25600/60000 (43%)]     Loss: 0.056873
Train Epoch: 44 [38400/60000 (64%)]     Loss: 0.054420
Train Epoch: 44 [51200/60000 (85%)]     Loss: 0.044693
Train Epoch: 44 Average loss: 0.051147

Test set: Average loss: 40.9841

Train Epoch: 45 [0/60000 (0%)]  Loss: 0.052639
Train Epoch: 45 [12800/60000 (21%)]     Loss: 0.047450
Train Epoch: 45 [25600/60000 (43%)]     Loss: 0.049975
Train Epoch: 45 [38400/60000 (64%)]     Loss: 0.052085
Train Epoch: 45 [51200/60000 (85%)]     Loss: 0.052505
Train Epoch: 45 Average loss: 0.051106

Test set: Average loss: 40.9865

Train Epoch: 46 [0/60000 (0%)]  Loss: 0.049335
Train Epoch: 46 [12800/60000 (21%)]     Loss: 0.054117
Train Epoch: 46 [25600/60000 (43%)]     Loss: 0.051621
Train Epoch: 46 [38400/60000 (64%)]     Loss: 0.048220
Train Epoch: 46 [51200/60000 (85%)]     Loss: 0.049373
Train Epoch: 46 Average loss: 0.051083

Test set: Average loss: 40.8203
```

```
Train Epoch: 47 [0/60000 (0%)]  Loss: 0.055486
Train Epoch: 47 [12800/60000 (21%)]     Loss: 0.053446
Train Epoch: 47 [25600/60000 (43%)]     Loss: 0.054560
Train Epoch: 47 [38400/60000 (64%)]     Loss: 0.052867
Train Epoch: 47 [51200/60000 (85%)]     Loss: 0.056047
Train Epoch: 47 Average loss: 0.050901

Test set: Average loss: 40.9508

Train Epoch: 48 [0/60000 (0%)]  Loss: 0.052556
Train Epoch: 48 [12800/60000 (21%)]     Loss: 0.049265
Train Epoch: 48 [25600/60000 (43%)]     Loss: 0.053140
Train Epoch: 48 [38400/60000 (64%)]     Loss: 0.053230
Train Epoch: 48 [51200/60000 (85%)]     Loss: 0.047147
Train Epoch: 48 Average loss: 0.050802

Test set: Average loss: 40.7031

Train Epoch: 49 [0/60000 (0%)]  Loss: 0.047615
Train Epoch: 49 [12800/60000 (21%)]     Loss: 0.050681
Train Epoch: 49 [25600/60000 (43%)]     Loss: 0.054231
Train Epoch: 49 [38400/60000 (64%)]     Loss: 0.049439
Train Epoch: 49 [51200/60000 (85%)]     Loss: 0.049550
Train Epoch: 49 Average loss: 0.050809

Test set: Average loss: 41.0263

Train Epoch: 50 [0/60000 (0%)]  Loss: 0.050358
Train Epoch: 50 [12800/60000 (21%)]     Loss: 0.051763
Train Epoch: 50 [25600/60000 (43%)]     Loss: 0.048958
Train Epoch: 50 [38400/60000 (64%)]     Loss: 0.051528
Train Epoch: 50 [51200/60000 (85%)]     Loss: 0.050002
Train Epoch: 50 Average loss: 0.050710
```

Reconstruction



Target



```
Test set: Average loss: 40.6148

The best (lowest) achieved loss: 40.615
```

Task 3 Written Answer

TODO 3.4: Compare the convolutional autoencoder to the fully connected autoencoder in 2-3 sentences. Mention both loss and visual reconstruction quality.

Your answer: The convolutional autoencoder has a signigicantly lower average loss of 40.6148 than the fully connected autoencoder with a average loss of 59.9906. Furthermore the conv autoencoder shows small visual improvements like the beginings of the logo on the white shirt in the upper left corner.